

CUSTOMMAPS: A NAVIGATION APP TO ADDRESS DISTRACTED DRIVING

by

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A Senior Honors Thesis Submitted to the Faculty of

The University of Utah

In Partial Fulfillment of the Requirements for the

Honors Degree in Bachelor of Science

In

Computer Science

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April 2025

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## Abstract

As technology continues to offer more features to ease the challenges of life, different considerations are presented. Navigation-based applications have become a cornerstone of everyday life for a significant number of people. However, the temptation and even need to use a mobile device while driving presents risks of its own. Distracted driving is a danger to the driver, others on the roads, and pedestrians. For example, in the state of Louisiana from 2016 to 2021, distracted driving accounts for a fifth of all fatalities. In an effort to address this, the CustomMaps application provides users with a variety of features, including the ability to draw custom routes and receive directions, either driving or walking, for this path, along with the ability to access Spotify from within the app to reduce how much a driver has to switch between apps on their phone while driving. CustomMaps also provides the typical features one would expect to find in a navigation-based application, including searching for destinations and receiving information on travel times and speed limits. Across the span of seven months, our team of four has worked in React Native to develop this cross-platform mobile app to give drivers more control over their routes and reduce distractions while driving. Having these customizations set up before beginning a drive allows the user to be more confident in the directions they are receiving, ultimately allowing them to place more attention on the road.

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## CHAPTER

### 1

#### Introduction

We've all been there: you're driving, following the navigation from your maps app, and it's telling you to turn, but you're unsure if this is the right place, since it doesn't seem to be the way you usually go. How do you check the route without removing your attention from the road? Another common scenario is when you're trying to change songs, and a dangerous situation arises as someone swerves in front of you while you are switching apps from your music app back to your maps app. This margin of time where a driver's attention is taken away from the road and directed to a mobile device presents danger to the driver and others on the road. Despite efforts to mitigate its impact, mobile devices and screens at large pose a distraction that is largely normalized and becoming more and more common in vehicles. While navigation apps have become a cornerstone for drivers, they pose an inherent distraction. This thesis aims to discuss the state of current maps apps and the dangers posed by mobile device usage while driving, while presenting a customizable maps application that allows users to set up their preferred routes before beginning a drive, as well as a music interface embedded in the app, leading to more confidence and less double checking during driving as well as limiting the need for drivers to switch between music app and navigation app while driving. This app serves as an all-in-one wrapper for driving interfaces, encouraging users to place more attention on the roads.



## CHAPTER

### 2

#### Related Work

##### **Current Navigation-Based Applications**

Map routing applications have become invaluable to aid in navigation for drivers, bicycle riders, and pedestrians alike. They offer real-time traffic information, hazard updates, and travel times using the user's location via GPS, allowing the apps to provide directions to the user as they drive. Many features continue to be added to cater towards the users' needs. Google Maps was released in 2005, making this technology still fairly recent and continuously evolving (Rădăcină et al.).

Maps have many uses, primarily navigation, but they are also used by lawyers to show locations in trials or other legal matters, where maps-based apps allow lawyers to take advantage of built-in features like access to Google Street View (Mighell 28). Mapping applications continue to evolve to provide as much information as possible to the user.

These apps can be useful even if you know where you're going, because you gain information about traffic to take a faster route. You can also save specific places, like your home address, to easily get directions to these pins. Business information is also a big feature, allowing users to see ratings and reviews for businesses located at the pin they select (Mighell 28).

Various GPS apps currently exist, but each have distinct issues they aim to address. Google and Apple Maps are focused on navigation, suggesting efficient routes. Waze uses crowdsourcing to provide real-time traffic, hazard, and police updates. Apple CarPlay allows for drivers to display their navigation and music in the same dashboard, but not all cars have the software for CarPlay, which limits the scope to which it can be used.

Waze provides a feature set catered towards private vehicle operators, such as road classification, detailing whether a road is a highway, county road, or other type, car lanes, speed limits, and parking lots. They allow their users to share information like police and hazards, and it learns paths that the user takes to suggest on other users' routes. However, it does not support navigation for public transit nor bicycles (Rădăcină et al.). Waze uses crowdsourcing to provide traffic reports of speed traps, accidents, road closures, and other hazards and things to be aware of when driving that constantly change and may not be predictable otherwise (Mighell 28).

Apple Maps uses vector graphics from open source map data, allowing it to use less data than Google Maps. Each application has a unique algorithm for route recommendation, which is revealed in the varying times and distances in suggested routes between the applications when run on the same start and end points. Interestingly, the same routes calculated on mobile and desktop were different in both estimated time and distance, highlighting the factors that the algorithms take into consideration. Only Apple Maps provided the same main route. The algorithms must take into account that to the human perspective, the shortest path is the one with the fewest turns, even if the distance traveled is larger (Rădăcină et al.).

In terms of features, Google Maps and Apple Maps provide a "leave at" service, while Waze does not. This allows the user to look at travel times for a different departure time than the current moment. However, while Waze and Apple Maps provide incident reports, Google Maps has limited reporting of incidents (Rădăcină et al.). All three do provide traffic status and location photos. Additionally, searching by different categories is another great feature all three apps provide to their users (Mighell 28). Google Maps clocks in with the highest number of features, while Apple Maps is ideal for iOS users (Rădăcină et al.).

In terms of visual appearance, the Google Maps UI provides a contrast balance with light colors. Waze's route suggestion provides markers for traffic information and hazards, but the color of alternate routes can be a bit difficult to see as they are drawn in a very light purple.

Another useful feature that Apple Maps and Google Maps allow is for users to avoid tolls and highways. Additionally, Google Maps offers fuel efficient routes that give information on steepness and type of roads. Apple Maps provides a brief description tag for routes it suggests - fastest route, fewer turns. In comparison to this, Waze has a lot of information in the user interface, which can make it hard and unintuitive to use as users may not be able to find the features or settings they are looking for (Rădăcină et al.).

Intelligent map agents can be used to learn the user's needs to better serve the user with the tailored geographic services they may need. These agents can act autonomously to achieve a task for the user, specifically in this case as a type of tour guide (Lanthier et al.). This shows that as technology continues to advance, so do our mapping applications. First it learns a location-based context, along with the user's interests, and history of visited places, as well as environment variables like time of day. Agents are also used to limit multitasking of a user, since an agent can tackle some of the tasks that a driver may focus on otherwise. The architecture of the system is organized as a collection of services, which communicate with each other by sending data and coordinating between services to achieve some task. These tasks include finding the shortest path cost-wise, in terms of distance and travel time, which use some complex algorithms to work on various types of terrains and model the geographical structure. There is also personal navigation, giving the user multiple options to choose from (Lanthier et al.).

## **Feature Preferences**

In a survey of 115 people, 23 marked integrating media applications such as Spotify as a favorite feature. 33 indicated directions being given at a wrong time, too soon/late/often, as a disliked feature (Rădăcină et al.). A main advantage of navigation apps is confidence in reaching the destination in the fastest way, as well as avoiding crowded roads, which is supported by the unique routing algorithms used to suggest routes to users. 25 noted focusing on the app while driving as a disadvantage (Rădăcină et al.). Overall, this gives insight into key points for any navigation app - it must have all the common, expected features while also presenting a specific and unique set of features to act as a point of attraction. It is worth noting that none of the above apps enforce app flow that addresses distracted driving, since the inherent use of a screen implies distraction, and there is no current link of music into a navigation app besides in Apple CarPlay which is not available to all drivers.

## **Distracted Driving**

Distracted driving poses a danger that has become quite normalized in the current day. It takes many forms, but all share the key idea of the driver having their attention diverted away from driving and towards a different activity. Distracted driving does not just pose a danger to the driver, but also to pedestrians and bicyclists who may be killed by distracted drivers.

Limited research exists to support the effectiveness of current efforts to address this danger, and the studies conducted tend to be based in driving simulators with limited metrics used to measure effectivity. However, the research done on the dangers of distracted driving is not to be ignored.

Some efforts to reduce distracted driving are in-app features which are free, while others are in hardware and can be over \$150. Apart from improving the infrastructure of the technology, other efforts to reduce distracted driving are based in policy enforcement and data collection. While around half the states in the United States have a cell phone ban in effect for drivers, this measure has mixed results (Mitran et al.). A part of this is likely due to the lack of metrics to measure distracted driving, because crash data can lack information, and other observational measures lack reliability.

One overview of the statistics backing the dangers of distracted driving was pooled from data from 2016 to 2021 in Louisiana, where they found that distracted driving fatalities accounted for about 20 percent of all fatalities in the state (Mitran et al.). However, the lack of concrete data collection metrics leads to uncertainty on the accuracy. Much of this data was collected via police accident reports or during crash investigation, which can be inconclusive.

Engagement in secondary tasks while driving has been shown to lead to decreased driving performance and increased risk of crashing. These secondary activities take many forms. It can be reaching over into another area of the vehicle, checking in on children in the backseat, changing the radio or music, or any other variety of task besides focusing on the road. Based on survey data from 6,002 drivers across the US in 2010, most drivers indicated that driving becomes more dangerous if they take their attention off of the road for more than 2 seconds. 65% of participants said that adjusting their music was a form of distraction (Mitran et al.).

A large portion of crashes are caused by distracted driving. In a study by Mitran et al., 1,691 moderate-to-severe crashes were reviewed involving young drivers ages 16 to 19, in the scope of single vehicle and vehicle to vehicle crashes with no fatalities. They found that up to 58 percent of these crashes were caused by distracting behaviors, and that cell phone use was more

common for road departure crashes than any other type of crash and contributed to significantly longer reaction times, averaging around 2.8s compared to the 2.1s reaction time of drivers who were not distracted. This stems from the observation that, for road departure crashes, drivers who had their attention diverted from the road to their phone before the crash did so for about four seconds, whereas drivers who were only listening or talking on the phone had an eyes-off-road time of 0.9s (Mitran et al.).

### **Comparison to Driving Intoxicated**

Using a driving simulator, cell phone drivers were compared to drivers at a blood alcohol concentration of 0.08% weight/volume, which is the legal limit. Both groups were subject to the same driving conditions. From this, they found that drivers talking on either a handheld or hands-free cell phone had delayed braking reactions and were involved in more traffic accidents than if they weren't talking on a phone. However, drivers intoxicated from ethanol drove more aggressively, tailgating the drivers ahead of them and braking more forcefully. This comparison led to the conclusion that cell phone driving and intoxication result in similar impairment when driving, and the dangers of distracted driving should be treated as seriously as drunk driving (Strayer et al.).

The rapid advance of technology-based distractions has made distracted driving a major problem, yielding the attention of many states which are emphasizing distracted driving and implementing more strategies to improve road safety (Mitran et al.). Use of spatial information has become a byproduct of our shift to reliance on technology and the use of traditional maps (Lanthier et al.). This has led to a greater dependency on spatial data and the manipulation of that data via apps and their algorithms, and the need for this is only growing further.

## CHAPTER

### 3

#### Capstone Overview

#### **Technical Overview**

##### **Tech Stack**

In this project, JavaScript and React Native were used to develop a cross-platform application. Node.js was leveraged for the runtime environment in conjunction with Expo Go to view the app. For the database, MySQL relational database management system was used. The server was run on an AWS EC2 instance with an S3 bucket for general storage of data.

The project leveraged multiple different APIs. An API is another sort of bundle of code that can be called within another project using an http request. It is like a machine that only responds to predefined questions. The Google Maps API involves a Nearest Roads API call which allows a placed point to adjust to the nearest road, as well as a Snap To Roads API call which allows a series of points to be adjusted to the nearest path on a road. Then, this is sent through the Directions API to get navigation directions for that path. Once this data is parsed, it is usable to display on the map and iterate through the navigation steps. The Spotify API was used to allow the user to login to their account, granting access to playlists from which songs can then be played.

GitLab was used as a project management tool and a CI/CD pipeline was set up to run a unit test suite on merges. A CI/CD pipeline is a tool for facilitating software updates by running certain steps, like building, testing, and deploying an application. Build stage is where the code is compiled and necessary libraries, or pre-existing modules of code, are updated as dependencies

of the project. This ensures high quality code as every stage and all tests must pass before the code can be merged into the main branch.

## Collaboration Practices

Throughout the project, an Agile work environment was maintained, with sprints running weekly Wednesday to Wednesday. GitLab issues were created for each feature and bug, as well as merge requests being made and reviewed by at least one other member of the team. The state of the project and goals for the week were discussed in weekly meetings at the beginning of each sprint.

## System Architecture Diagram

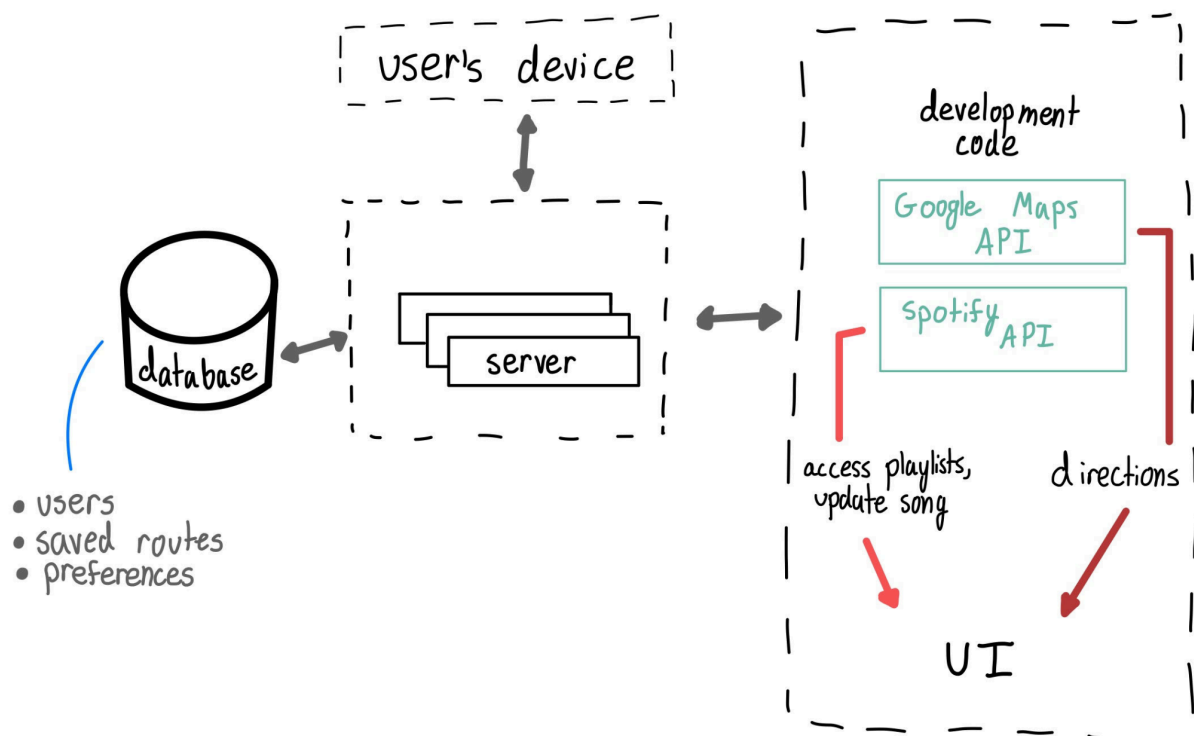


Figure 1. System Architecture Diagram of the CustomMaps app.



In figure 1, it is seen how the user's device sends information to the server, which is stored in the database and also updates the User Interface (UI). The development code, which includes the Google Maps API and Spotify API, can provide directions and provide music features like changing songs and playlists, which then is sent back through the server to the user's device. The database contains multiple different tables linked via the user ID assigned to each unique user, which stores the preferences and saved routes.

## **Project Overview**

The aim of this project was to create an all-in-one navigation app that catered to the user's needs, providing various customizations and preference options for the user to change to their liking. This addressed the limitations of current apps, which have a tendency to limit customization and are rather one-size-fits-all. This ties into one of the main features of the app, the customizable drawing portion, which allows users to draw out their path and then request directions for that route. Additional features include saving routes, which allows users to save and then retake routes for efficient flow of use, as well as Spotify integration, which allows users to login to their Spotify account and select songs and playlists from an interface within the app, removing the need to switch apps as the mini player overlays the map screen.

Main features of the app also include providing navigation while the user is driving or walking, along with speed limit information and a time estimate for the route. This ensured our product was still functional for typical GPS users, with most expected features present and intuitive. This also encompasses an expandable search bar for a source and destination, and the ability to switch to walking mode to receive a path and directions for walking, which shows the elevation changes of the path encoded in colors.

## Individual Contributions

My contributions to the project involved various technical portions as well as the custom drawing feature. I initially set up the project through Expo Go and React Native, making the decision that JavaScript would be the best language for the project. After setting up the files in GitLab, I loaded in the map background and set up a request for location permission from the user to then display their location on the map. My main focus for the prototype was to create a proof-of-concept demo of the custom drawing feature, which involved dropping pins on the map and selecting a destination, which then would return directions to the destination going through the pins. While this allowed for a simpler backend, it limited the functionality of catering to the user's intention. It also allowed for silly paths to be given to the user due to the algorithm used for quickest route calculations.

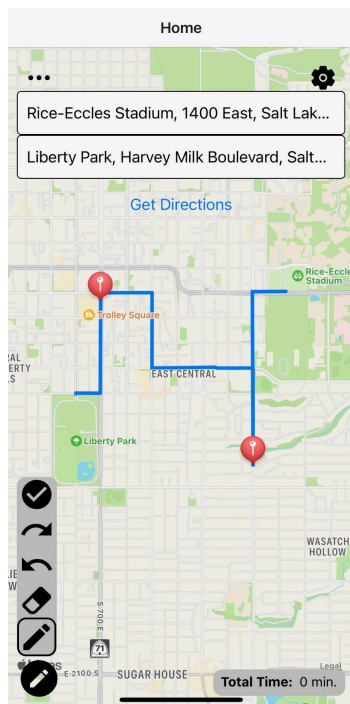


Figure 2. Prototype custom drawing feature with the route navigating through two additional points.

To pivot from this, I determined it would be best to allow the user to fully draw an entire route from start to finish and then request directions on that, rather than slightly modifying the already given route. This was also necessary since the Google Maps API does not provide road intersection data, which limits the ability to select portions of the road without adding complex logic. Ultimately this pivot worked well for the scope of the project. This allows the user to draw the precise path they would like to take while still getting the real-time navigation assistance a typical maps app would provide.

To expand out the drawing feature, I added undo and redo functionality, along with the option to trash a drawn route and start over. If selected, it prompts the user to confirm they want to erase the route, as seen in figure 3, giving them the option to say no and keep the current drawn section. Then, once the user has the drawn route they like, they can confirm the route and get directions.

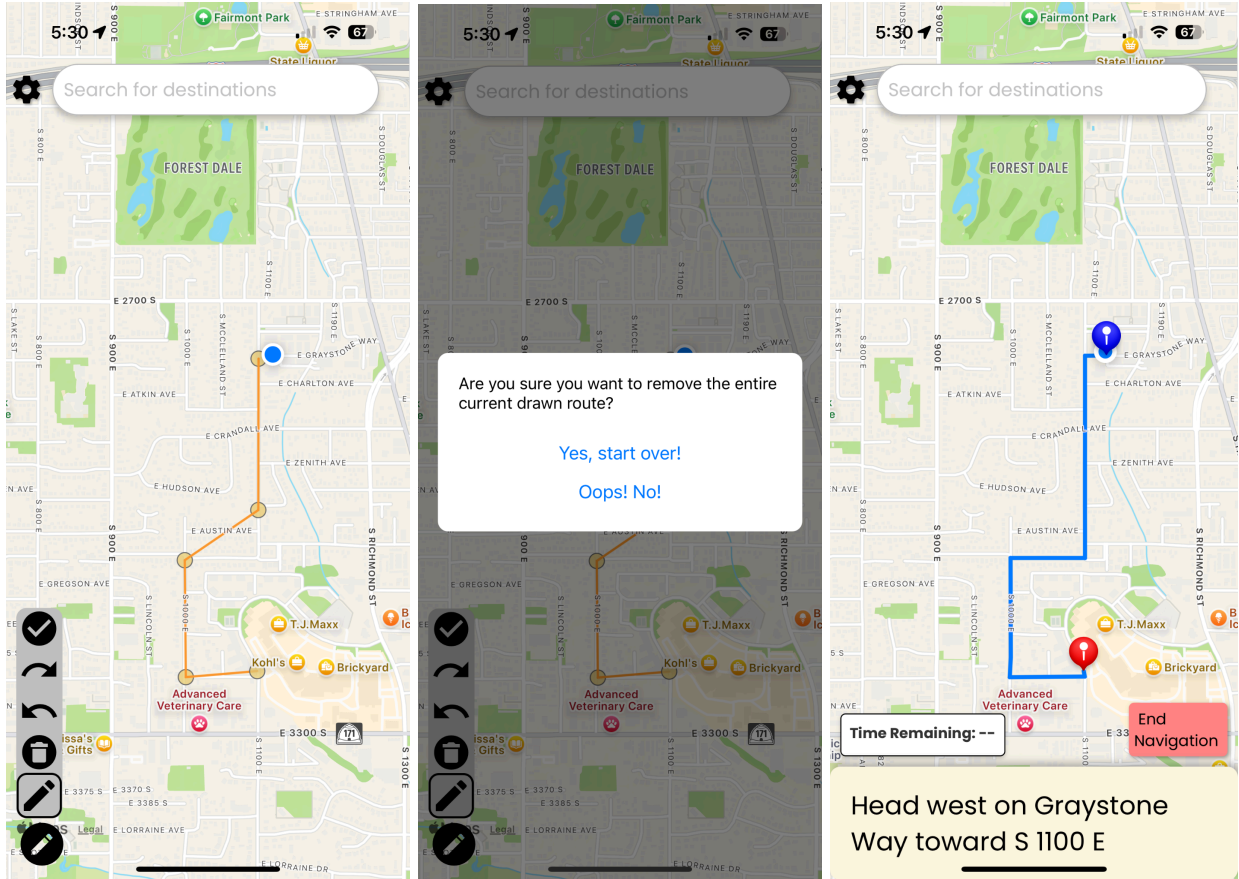


Figure 3. Final custom drawing flow with points dropped, confirmation for deletion modal, and the given path that goes through the path the user drew with directions given.

After shifting my contributions to tidy up the user interface, I also implemented walking mode to give the user walking directions and path, as well as fixing the implementation of the end navigation button so that it properly reset all variables and all of the map components. Then, I worked on setting up the CI/CD pipeline, which proved to be quite difficult as I have not had any real experience configuring this type of technical component. Additionally, since we were not running our app inside a container like Docker, it made setting up the pipeline particularly difficult since we are all running different operating systems. Our workaround for this was for me to configure the pipeline to run locally on my machine, and then I would run the pipeline

whenever someone made a merge request. This worked for the scope of this project and was a learning opportunity to see what I would do differently in a similar project setup - namely, using a container like Docker and setting up the pipeline and unit tests early on in the process rather than as an afterthought.

Upon getting the pipeline configured properly and running Jest, a unit test package, I worked on creating an expansive unit test suite to ensure our functions worked as expected.

I then added minor bug fixes with our navigation portion, along with the option for users to login as a guest without an account to still have quick access to the app but without the ability to save preferences.

## User Controls

Upon opening the app, the user is greeted with a Login screen, where they can create an account or log in to an existing account. If they wish to proceed without an account, they are able to do so as well.

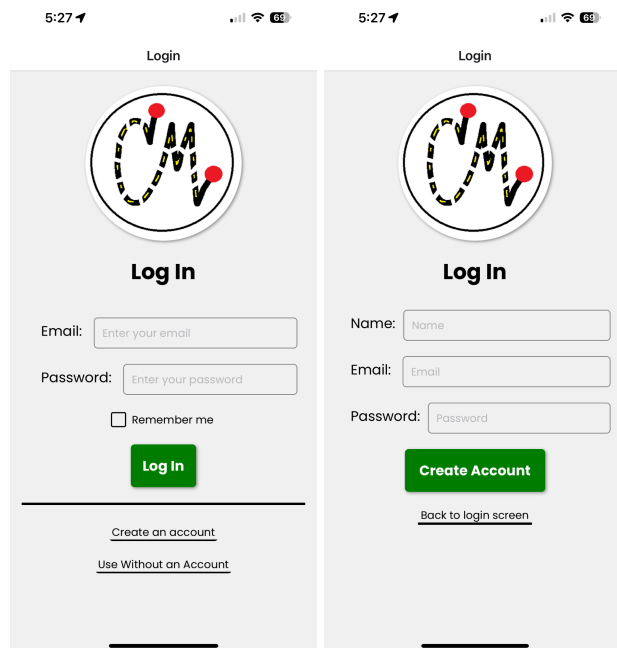


Figure 4. Login screen with options to create an account, pictured right, or use without an account.

Then, the user is brought to the main screen, with their location displayed in the center, as seen in figure 5. They can click into the search bar and search for a destination, which expands the search bar.

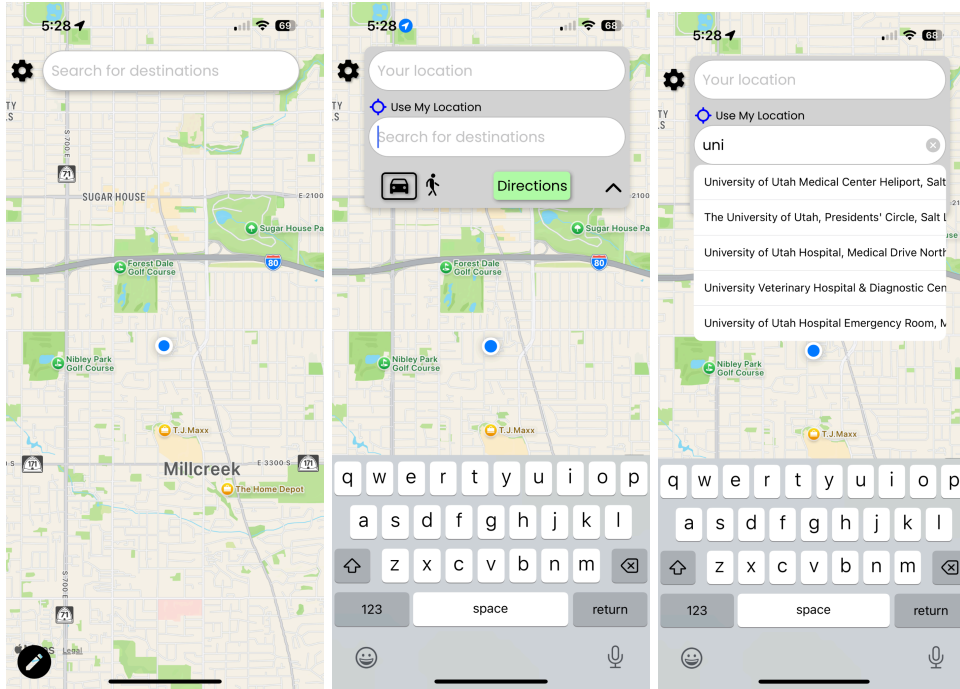


Figure 5. The main map screen with the expandable search bar and auto-complete search functionality along with walking mode option.

Upon selecting a destination and clicking directions, the user sees the screen in figure 6, displaying in text the direction which is also read aloud if they have sound enabled. The speed limit and total time remaining is displayed as well, ensuring the user has all of the information they may need displayed on the same screen.

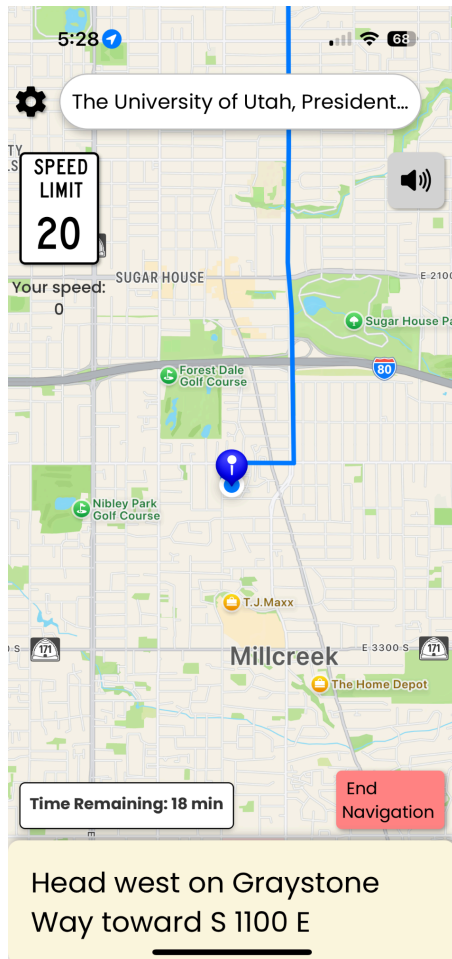


Figure 6. Navigation is presented to the user with information including next direction, speed limit, total time remaining, the option to end navigation, and the option to mute the directions being read aloud.

If the user wishes to further customize the app, they can navigate into the preferences screen (see figure 7) and select from any of the dropdowns the features they would like to change.



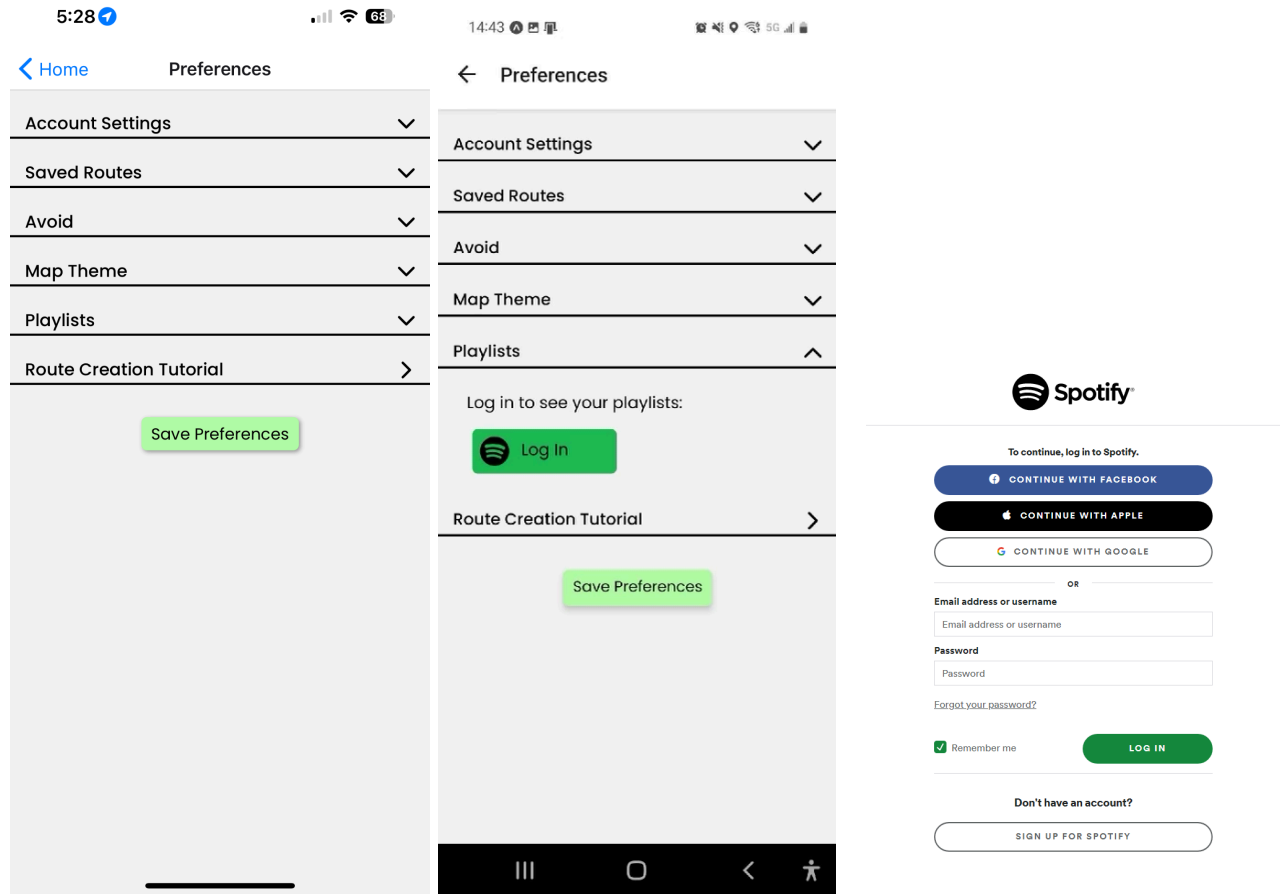


Figure 7. Preferences screen and logging into Spotify.

In the preferences menu, the user can log in to their Spotify account under the Playlists tab (see figure 7). After logging in, they have the ability to select any of their playlists and change the song (see figure 8). Then, when they navigate back to the main map screen, the user is greeted with the current playing song and the ability to skip, pause, or go backwards. This stays even while the user is driving, giving them the option to control their music from within the navigation app.

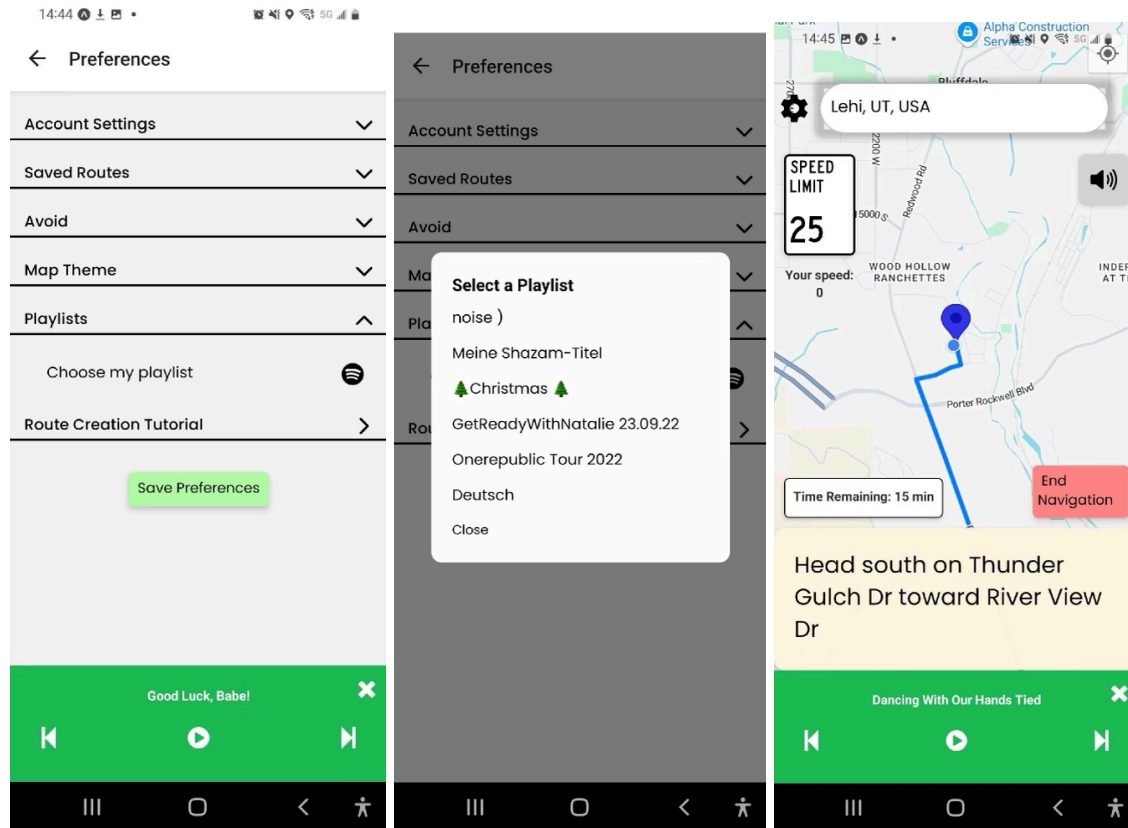


Figure 8. After the user logs in, they can select a playlist from within the app and play or change the song.

If the user decides to make a custom route, they can open the drawing panel and begin dropping pins which appear as orange dots, which then connect themselves to each other and display a connected line. In this menu, the user has the option to undo or redo points they place. If they decide they want to erase the entire route, they can click on the trash bin icon and are prompted with a modal to confirm deletion (see figure 9).

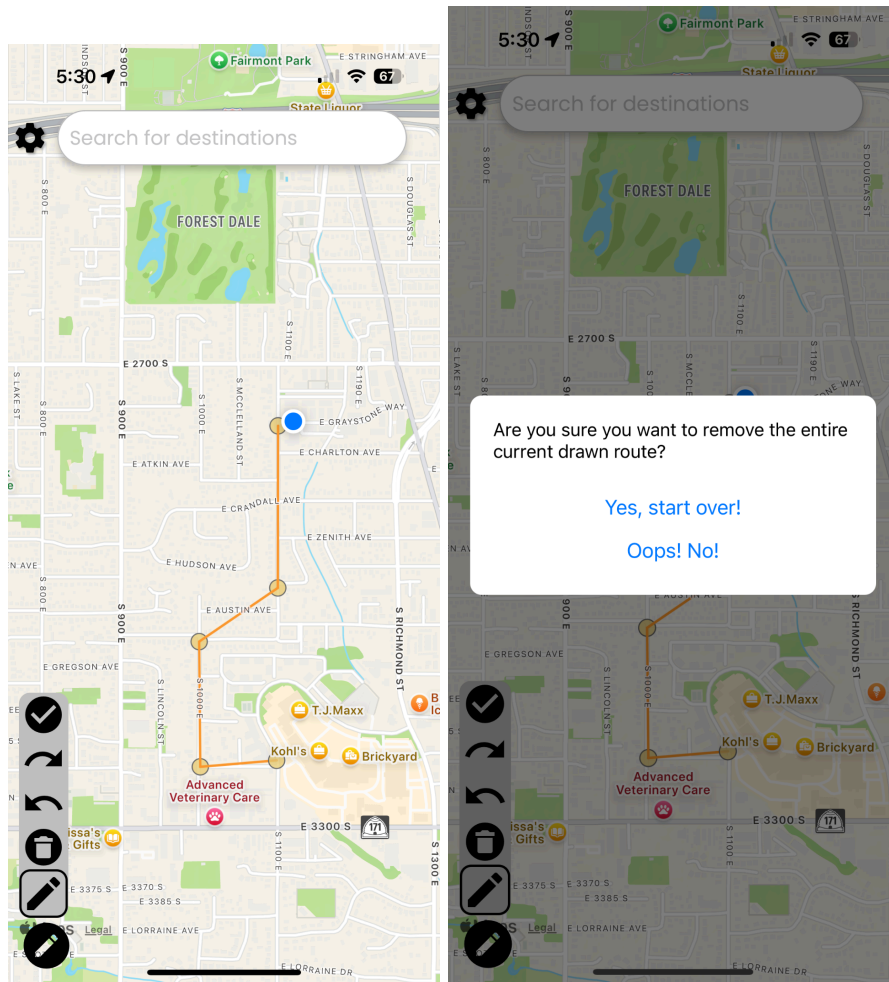


Figure 9. Once the drawing menu is open, the user can place points which connect into a custom route. They can undo or redo points, as well as delete the whole path. Delete prompts a confirmation modal.

Once the user has a path they like, they can click the checkmark, and they will receive directions for that path. After a user completes a drive by clicking end navigation, they are prompted to enter a name for it and save the route.

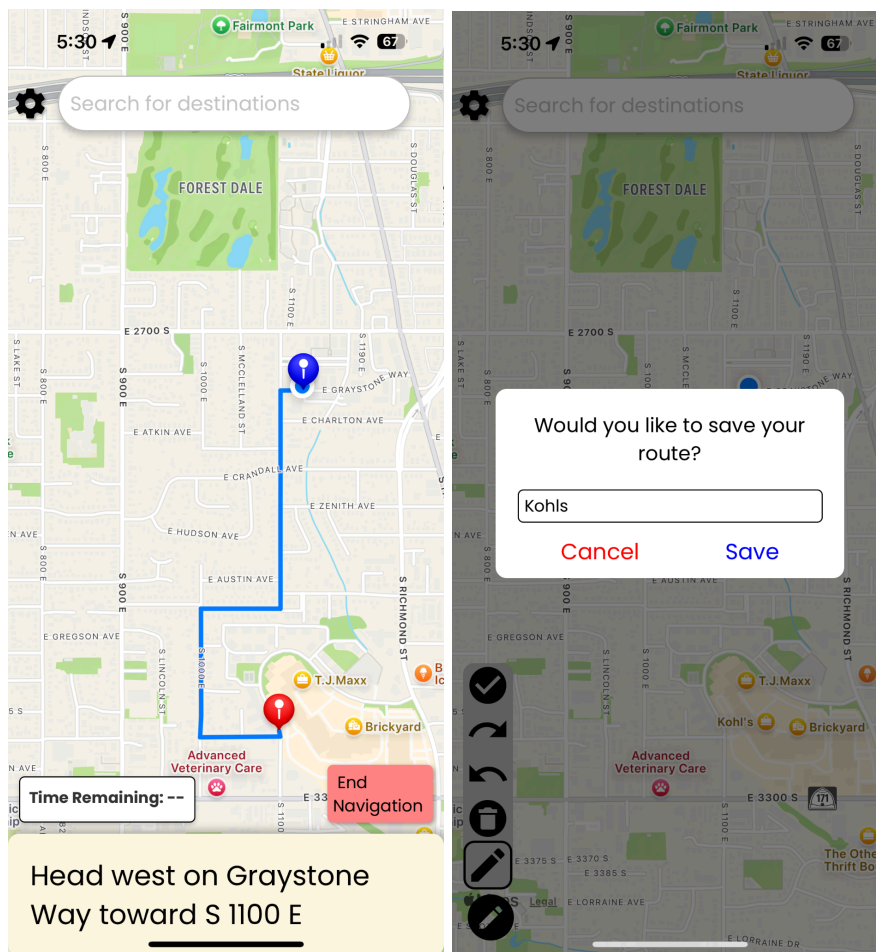


Figure 10. The user confirms the route and gets directions along the path they have marked. On finishing a drive or walk, the user can save the route and give it a name to take another time.

## CHAPTER

### 4

#### Addressing Distracted Driving within the Capstone Project and Future Directions

Distractions appear in many places when driving, and while current navigation apps do attempt to address this issue, there is still a large area for improvement. Distraction as a factor for traffic fatalities and injuries has increased nationally in recent years, making distracted driving a priority to be addressed for many states' transportation agencies and other stakeholders in regard to safety (Mitran et al.). One idea for reducing distracted driving is to promote smartphone technologies to minimize distracted driving, which is what CustomMaps attempts to do. Our app is an in-app way to address distracted driving.

As previously noted, 65% of participants in the survey from Mitran et al. indicated that adjusting their music was a form of distraction. This became a focal point for one of our main features which involved integrating Spotify into our app.

#### **Features to Prevent Distractions**

##### **Create a Route**

One place of distraction comes from checking a route during navigation, since increased reliance on an app's navigation has led experienced drivers to question their own navigation. If something looks wrong to them but is right according to the app, it can fluster the driver and take their attention off of the road. In this case, the user's attention is drawn to the screen. To address this, we allow users to create their own routes while still getting the navigation assistance for upcoming turns and directions. This ensures the user is never second guessing provided

directions because they know they are taking the route they want to take, so they can focus more on the road.

For this functionality, the user is able to draw out the exact route they want to take beforehand by placing points on the map which connect to each other. These points snap to the nearest road, mimicking drawing on a real map. The user is able to undo or redo changes, or trash their route and start over. Once the user has a route drawn that they are satisfied with, they can request directions, which ensures the route is valid. Valid here indicates that, for example, they are not going down a one-way road the wrong way. If they did indicate this, the path they are given would be routed around that point to the closest valid road. Then, navigation is provided while the user is driving with both readable directions at the bottom of the screen, which is helpful for a copilot, and with the directions being read aloud.

## **Spotify Integration**

Another palace of distraction comes from drivers switching between apps, which takes attention away from the road. In this case, users would normally direct their attention to their mobile device to switch apps and then change their music, then switch back to their mapping application, which takes quite a bit of time and is a significant source of distraction. Drivers may switch between music apps like Spotify and their navigation apps on the road frequently. To address this, we integrated Spotify to be within our app, so the user does not have to navigate away to change their music and can instead place more of their attention on the road.

To ensure ease of use, the user can navigate into the Preferences Screen and login to their Spotify account before beginning their drive. If logged into their user account, they can indicate their Spotify to be remembered and only have to login once. Then, the user can select a playlist

from their list of playlists right within our app. Once navigated back to the main screen, the user is greeted with the current playing song and buttons to pause, skip, or go backwards a song. This mini player is located at the bottom of the map screen, meaning the user does not even have to change screens from within our app to adjust their music.

## **Future Directions**

One place of distraction that was left unaddressed by our app was messaging while driving. A possible future direction could be to block notifications or allow the user to toggle on a read aloud feature so the driver does not have any reason to look at and use their device for texting.

Another point of interest is the lack of eyes-off signals drivers rely on to adjust settings in their car. Knobs give the user a gauge of how much they are changing something and allow drivers to make adjustments to the radio, heat, and other settings while keeping their eyes on the road by using the sensory input from the knob or feedback from a button. However, with screens, drivers must divert their attention away from the road and to the screen in order to know what they are changing, which also takes more time. Interestingly, Volkswagen has observed this and has begun going back to knobs and buttons (Schwarz), in contrast to many other vehicle brands that have begun installing huge screens in place of the dashboard. One way to integrate this idea into our app would be that, when driving, a touch on the app's main screen registers as a pause music signal and vibrates the phone, and a double-tap would skip a song with similar feedback from the device. Additionally, a different direction could be to link the functionality of the app into the dashboard controls to be able to make the same radio adjustments with the knobs and have those signals be sent back to the app.

## CHAPTER

### 5

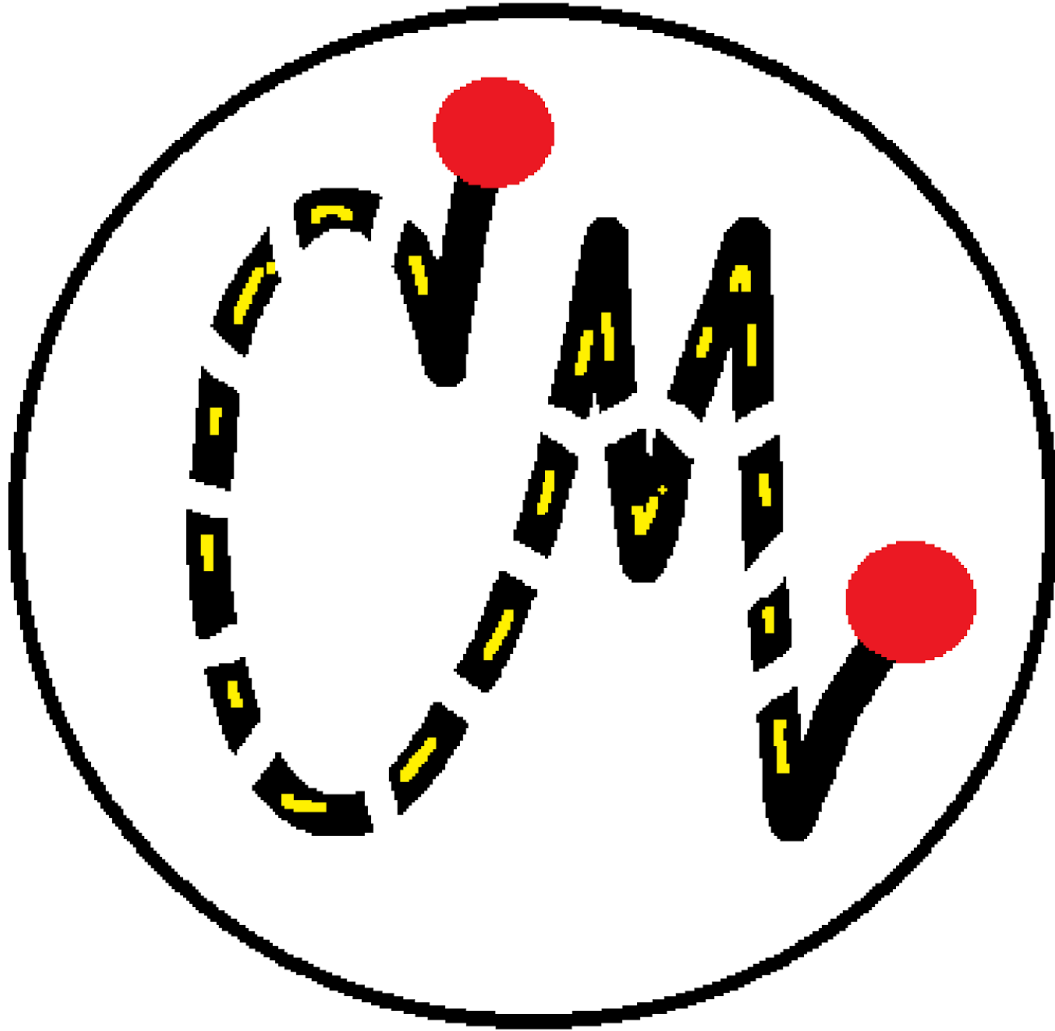
#### Conclusion

Navigation-based applications have become an important part of most people's everyday routine. However, they can lead to distractions in the form of encouraging phone usage while driving to check a route or change music, along with many other tasks like phone calls or answering texts. A cross-platform navigation app built in React Native, CustomMaps aims to address this by integrating a customizable drawing feature so users can ensure they are on the roads they want to be on, along with integrating Spotify to allow users to have control of their music without needing to change apps. Allowing users to have more control over their routes ensures the driver can place more of their attention on the road, since distracted driving has become an increasingly important problem to be addressed.



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# CustomMaps

Senior Capstone Design Document

Jessie Taubert, Kira Halls, Natalie Wilkins, Nate Stovak

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# 1. Executive Summary

CustomMaps is an innovative mobile application designed to give drivers full control over their routes while still benefiting from the essential features of modern navigation apps. Unlike existing map apps that suggest the quickest or most efficient route, CustomMaps empowers users to customize their paths, whether they want to avoid specific roads, add stops, or simply prefer a particular route due to familiarity. Our app eliminates the hassle of "tricking" navigation apps by dropping pins or adjusting settings to fit a preferred route.

Our app addresses a common frustration among drivers who have a preferred way to reach their destination but need navigation assistance. It provides users with a seamless way to select their desired roads, either from scratch or by modifying a suggested route. Through an intuitive interface, users can click or drag to adjust sections of their route, avoiding roads they dislike or adding necessary stops. The system ensures safety by preventing users from selecting dangerous or illegal routes, such as wrong-way entries on one-way streets.

CustomMaps offers a range of useful features, allowing drivers to design their own paths while still receiving real-time updates on hazards, police presence, and estimated time of arrival (ETA). Our app also allows users to save routes so they can quickly take a previous route. Machine learning integration will analyze previous trips to predict the user's preferred routes and adjust future suggestions based on their likes and dislikes. Our app is built with React Native in JavaScript, ensuring it will be accessible across both Android and iOS platforms.

What makes CustomMaps particularly novel is its ability to combine personalized route customization with the core functionalities of existing navigation systems. By using machine learning to predict user preferences, it offers a tailored driving experience that not only guides users to their destination but does so in a way that aligns with their habits and preferences. CustomMaps fills a unique gap in the market, delivering stress-free navigation while giving drivers the freedom to take the routes they truly want to follow.

## 2. Background

### 2.1 - Current Related Solutions

While there are a number of established navigation apps like Google Maps, Apple Maps, and Waze that provide highly functional navigation tools, they often lack sufficient flexibility for users to personalize routes. Google Maps allows route manipulation to an extent, but users must manually drag routes, which can be tedious and imprecise. Waze offers real-time hazard and traffic alerts but prioritizes speed over customization, often rerouting drivers away from preferred roads.

Garmin Maps has one of the closest established solutions to our proposed solution through their adventurous routing. Their adventurous routing mode takes in the user's preferences of non-freeways, curvy roads, and hills, and creates future routes based on those preferences; however, it is only offered on Garmin-backed products and does not allow for as much route manipulation as we would implement. Our solution will have a similar user base and functionality but will be more accessible and allow for more route modifications by the user.

CustomMaps differentiates itself by placing user control at the forefront of its design. Our app allows for easy drag-and-drop route manipulation while also learning user preferences over time with machine learning algorithms. This ensures that users can not only create a personalized route but also benefit from predictive suggestions based on their previous trips. Additionally, CustomMaps focuses on an intuitive interface that allows for quick customization, saving time and effort compared to existing solutions.

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### 2.2 - Technologies

To create our mobile app, we will use JavaScript and ReactNative to polish our front end and tie together our APIs. We will be using Google Maps vector tiles for the map API, as well as TomTom's traffic API to find traffic backups, construction, and accidents on our proposed routes, along with the historical time it takes to get from one destination to another. This would allow for accurate ETAs and important safety information for users. Time permitting, we will also integrate the Spotify API to allow users to attach playlists to certain routes and potentially suggest playlists. Using these APIs will allow us to not recreate similar assets from scratch, as they already provide information and services our app requires.

The machine learning aspect of our app will be backed through a MySQL database, where we will store user preferences. This algorithm will be developed in Python using machine learning-based libraries in Python such as Pytorch, Pandas, Numpy, and Scikit-learn.

Our database will also store the users' saved routes for easy access for future navigation, preferred types of routes such as surface streets, preferences to avoid certain areas, etc., and songs/playlists used by users on specific routes. This data stored in the database will be mostly ordinal and categorical data, making the data relatively easy to encode for a logistic regression-based neural network to provide suggested routes to users in the future.

To provide optimal safety and convenience for users, we will cache the current, in-progress route in case the user enters a zone with no service, allowing the user to continue to receive navigation.

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## 2.3 - Software Requirements

The software/hardware requirements for CustomMaps are relatively straightforward. Users will need a smartphone running either Android or iOS operating systems. Our app will not require high-end hardware or extensive memory, as all machine learning processes and route computations will be handled on cloud-based servers. However, users will need mobile data and location services enabled on their devices to ensure accurate navigation, real-time traffic updates, and personalized route suggestions.

# 1. Requirements Analysis

## 3.1 - System Architecture

### **User Interface (UI)**

Description: Built with JavaScript and React Native, the mobile UI is the user-facing portion where drivers can set routes, adjust preferences, and interact with map features.

Interaction: Sends user actions/events (like route requests, preference changes, and route enjoyment) to the API Gateway that receives map, traffic, and route updates or the backend server depending on the specific event.

### **API Gateway**

Description: Acts as a central hub for managing external and internal API requests. Later sending this data to the backend server to interpret for UI usage

Third-Party APIs:

Google Cloud API: Fetches map tiles for UI rendering.

TomTom API: Provides real-time traffic data and hazard updates.

Spotify API: (Optional) Attaches playlists to routes for user experience

Open Elevation API: Shows elevation data for latitude and longitudinal points, allowing maps to show incline information for walking routes

Interaction: Receives requests from the UI, forwards them to external APIs, and manages responses back to the Backend Server for additional processing.

### **Backend Server**

Description: Handles business logic, data processing, and coordinates between the API Gateway, Machine Learning Module, and Database.

Subcomponents:

Route Customization Processor: Manages real-time route adjustments.

Data Handler: Manages data requests to and from the Database.

ML Integration: Connects to the Machine Learning Module for predictive routing.

Interaction: Deals as middleman between database and backend functions

### **Machine Learning Module (Python)**

Description: Uses deep neural networks to analyze stored routes and preferences to predict the user's preferred route. It will intuitively learn as the user continues to use the application and the database receives more data for that user.

### **Database (MySQL)**

Description: Stores user preferences, routes, and playlist associations, providing quick data retrieval. This data is further used to enhance the user experience to seamlessly create a personalized application. Utilizes server to work with backend.

## System Diagram

This is a baseline guide for communication between the UI, APIs, backend server, and databases as outlined above. User events include rating routes they've completed, creating a navigation path, beginning, taking, and ending a navigation session, and similar interactions with the UI display.

## 3.2 - Personnel

**Kira Halls:** Kira is a Computer Science major with experience in front-end development, particularly in designing intuitive user interfaces. She also has a strong programming background. As such, she will be responsible for:

User Interface: Working alongside Natalie Wilkins in designing/implementing the UI

API Gateway: Integrating the APIs to interact with the user interface and handling responses from the back end, specifically the main integrator of the Open Elevation API

Backend Server: Building the backend to handle business logic and data processing, and assist in coordinating between the database, API Gateway, and Machine Learning Module as needed.

**Natalie Wilkins:** Natalie Wilkins is a Computer Science major with full-stack internship experience. She also has an eye for designing effective UI's. She will be responsible for:

User Interface: Working alongside Kira Halls in designing and implementing the user interface

API Gateway: Integrating the APIs to interact with the user interface and handling responses from the back end, specifically the main integrator of the Spotify API

Backend Server: Helping interpolate data from the backend of the UI into the database

**Jessie Taubert:** Jessie Taubert is a Computer Science major with a strong programming background and has a strong understanding of how our application will operate. As such, she will be responsible for:

API Gateway: Integrating the APIs to interact with the user interface and particularly focusing on how the APIs handle responses from the back end.

Backend Server: Building the backend to handle business logic, data processing, and coordinating between the API Gateway, the Machine Learning Module, and the Database

**Nate Stovak:** Nate is a Data Science major with a background in machine learning. He also has taken the Databases course and has database experience. As such, he will be responsible for:

Machine Learning Module: Building the machine learning module and neural network to learn and analyze user's route preferences. Integrating the machine learning model with the backend server and the database to train and execute the model.

Database: Building, accessing, and maintaining the database to include user data in our application



### 3.3 - System Features

| Priority | Feature                                    | Feature Code | Responsible | Feature Description                                                                |
|----------|--------------------------------------------|--------------|-------------|------------------------------------------------------------------------------------|
| 1        | Begin navigation                           | M.1.1        | Jessie      | Button to begin navigation once it has been computed                               |
| 1        | Create route from scratch                  | M.1.2        | Jessie      | A user can draw lines on a map and be guided on that route.                        |
| 1        | Create route from search bar               | M.1.3        | Jessie      | The user can input a source and destination and be guided on that route.           |
| 1        | Display directions                         | M.1.4        | Natalie     | The directions for the user's selected route will be displayed on the screen.      |
| 1        | End navigation                             | M.1.5        | Jessie      | A user can choose to end navigation and have directions stopped.                   |
| 1        | Read directions aloud                      | M.1.6        | Natalie     | If the user toggles sound on, they will be read directions for their route.        |
| 3        | Add stop during navigation                 | M.3.1        | Kira        | The user can add an additional stop while on their route.                          |
| 3        | Create walking route                       | M.3.2        | Kira        | Routes similar to roads but utilizes walking routes and walking times              |
| 3        | Report an incident                         | M.3.3        | Natalie     | A user can report an incidents on their route                                      |
| 3        | Show route incline info                    | M.3.4        | Kira        | User can toggle walking mode to show elevation changes on routes                   |
| 3        | Toggle sound                               | M.3.5        | Natalie     | The user can toggle the sound, indicating if the directions would be read aloud    |
| 1        | Add specific road to list of avoided roads | P.1.1        | Nate        | The user can add specific roads to avoid during directions                         |
| 1        | Preference menu                            | P.1.2        | Kira        | The user will be able to open the preference menu to access different options.     |
| 2        | Scenic preferences                         | P.2.1        | Nate        | User can select scenic preferences to adjust routes based on user preference       |
| 2        | Select/start/recommend a playlist          | P.2.2        | Natalie     | The user can select and pre-made playlist and recommend playlists based on factors |
| 3        | Change map theme                           | P.3.1        | Kira        | The user can change the theme of their app, e.g. dark mode, topo mode, etc.        |

|   |                             |       |         |                                                                                    |
|---|-----------------------------|-------|---------|------------------------------------------------------------------------------------|
| 1 | Edit a saved route          | A.1.1 | Jessie  | The user can edit a route they have saved.                                         |
| 1 | Retake a saved route        | A.1.2 | Jessie  | The user can select one of their saved routes and navigate through it.             |
| 1 | Save a route                | A.1.3 | Jessie  | The user will be able to save a route so that they can use it again later.         |
| 1 | Saved Routes Menu           | A.1.4 | Natalie | The user can navigate to a menu with options related to saved routes.              |
| 2 | Copy and edit a saved route | A.2.1 | Jessie  | Allow the user to make a copy of a saved route and then edit it.                   |
| 2 | Create account/Login        | A.2.2 | Kira    | User can create account or login to save info                                      |
| 3 | Give a rating               | A.3.1 | Nate    | After a user completes a navigation, they will rate the route to understand users  |
| 1 | Settings menu               | E.1.1 | Natalie | The user will have options within the settings menu related to their account.      |
| 2 | Toggle saving data          | E.2.4 | Nate    | The user can decide whether they would like their data to be saved                 |
| 3 | Tutorial                    | T.3.1 | Kira    | Initial tutorial for new users introducing them to the features of our application |

# 1. Software Engineering Tools and Techniques

## 4.1 - Agile

Our team will use an Agile development cycle to allow for flexibility in our process. Each sprint will last two weeks, with check ins once a week to discuss progress made, roadblocks, etc. Our sprints will be planned out a few weeks in advance where possible, but the agile development process allows us to introduce changes as needed during each sprint.

## 4.2 - Versioning and Branching

Our team will use GitLab as our main tool for versioning and branching. Each team member will have their own branch, and will always commit to that separate branch. Once their section of code has been completed, the user will create a pull request to main which will be approved once at least one team member has reviewed the code.

## 4.3 - Bug Tracking

We will use GitLab Issue Boards to manage bug tracking, ensuring that any defects discovered during development are efficiently documented and resolved. Each bug will be logged, added to a dedicated "Bugs" list, and then moved through the workflow stages, similar to any other issue in our project. This approach will give our team clear visibility into each bug's status, helping us prioritize and systematically address issues as they arise. By leveraging GitLab's tools, we will streamline the bug tracking process, making it easier to stay organized and focused on maintaining code quality.

## 4.4 - Testing and Code Review

For testing and code review, we will establish a thorough process to ensure each feature meets our quality standards before being marked as complete. We will write test cases specifying the steps required to validate each feature, and other team members will replicate these steps to confirm the feature functions correctly. Code review requirements will vary by feature priority: Priority 1 features will need two reviewers to ensure thorough oversight, while Priority 2 and 3 features will need only one reviewer. We will rotate reviewers to encourage cross-team familiarity with different parts of the codebase. Once the code has passed review, a pull request will be initiated to merge the feature into the main branch, solidifying our collaborative and high-quality approach to development.

## 4.5 - Documentation

We plan to make our documentation centralized through gitlab. This includes updating our gitlab wiki page after code sprints and most likely during as well to keep up with developments of the project. Comments on pushing/pull requests to ensure that we can keep a timeline of events and revert back to any updates with ease if we happen to encounter a problem. Header comments in each file will be an essential part of our coding journey to ensure readability. These headers will be the roadmap to any given file making it fast and easy for another user to code review their teammate and know what the impact is of the file they are looking at.

## 4.6 - Team Communication and Meetings

Our team communication happens through text messages to create a fast-reply system. These messages are sent to each individual's phone which everyone has on them at any time, resulting in an environment where any query or statement is responded to efficiently. Our meetings happen primarily over zoom on Wednesdays and/or Fridays during class time. These meetings usually last 30 minutes to an hour allowing us to work together, ask questions in real time, and plan how we are contributing in the future. Our code sprints are going to be one week long, with room for changes once we realize how we all will feel comfortable. Each person gives an update on their work at least weekly.

## 4.7 - IDE and Libraries

The IDE we will be using is VS Code. We will be coding in JavaScript using the React Native library with Expo for mobile app development and Node.js. We will also be using Python and Python libraries

including Pandas, Scikit-learn, and Numpy for our machine learning algorithm. Our database will be built on MySQL.

## 4.8 - Issue Tracking

We plan to keep track of our issues and features using the Gitlab issue board and tagging them with the proper labels. When encountering major roadblocks, we will raise these issues to the team via the communication methods previously.

# 1. Timeline

|                            | Jessie                                                                                                         | Kira                                                                                  | Natalie                                                                                     | Nate                                                                                                   |
|----------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Week 1<br>Jan 6 - 12       | Debug current issues with source/destination selections                                                        | Creating a new account                                                                | Search bar features (switch order of source and destination, route options, avoid highways) | Creating global server through AWS, so it is not just local                                            |
| Week 2<br>Jan 13 - 19      | Change pins to be selecting and connecting road segments                                                       | Logging in with a Google account                                                      | Read directions aloud, make sure directions are displayed correctly, toggle sound           | Create user rating system after they reach their destination                                           |
| Week 3<br>Jan 20 - 26      | Algorithm for route selection (working on the "custom" portion), snapping road sections/ "building out" a path | Preferences menu functionality - changing map theme                                   | Follow user on map, display directions at correct time                                      | Create algorithm that takes into account road to avoid                                                 |
| Week 4<br>Jan 27 - Feb 2   | Debugging                                                                                                      | Bugs                                                                                  | Bugs                                                                                        | Debugging                                                                                              |
| Week 5<br>Feb 3 - 9        | Gather data from drive, storing roads to avoid                                                                 | Preferences menu functionality - avoid specific roads                                 | Add functionality to settings menu                                                          | Creating algorithm to learn based off users preferences                                                |
| Week 6<br>Feb 10 - 16      | Loading in saved routes                                                                                        | Adding accessibility elements to app (alt to images, confirming color contrast, etc.) | Roads to avoid                                                                              | Debug previous algorithm and start working on algorithm that will take into account scenic preferences |
| Week 7<br>Feb 17 - 23      | Implementing traffic API & travel times, displaying speed limit                                                | Dynamic adjustment of route time estimates using traffic and travel time              | Touch up navigation features                                                                | Finish algorithm that takes into account scenic preferences                                            |
| Week 8<br>Feb 24 - March 2 | Driving, creating routes, saving routes, all fully functional                                                  | Adding stops during navigation, finishing up preferences menu functionality           | Bugs                                                                                        | debugging                                                                                              |
| Week 9<br>March 3 - 9      | Implementing ML Model with the suggestions to routes                                                           | Walking mode and route elevation display                                              | Report an incident button                                                                   | Work on smooth transitions between phases ensuring fluid app                                           |
| Week 10                    | Adding stops in a                                                                                              | Create tutorial, polish                                                               | Implement Spotify                                                                           | Assist in implementing                                                                                 |

|                          |                                |                               |           |                                                                              |
|--------------------------|--------------------------------|-------------------------------|-----------|------------------------------------------------------------------------------|
| March 10 - 16            | route, swapping order of stops | up UI                         | playlists | spotify playlists with weather and previous playlists so it is based on mood |
| Week 11<br>March 17 - 23 | Transit mode / walking mode    | Buffer time, assist teammates | Bugs      | Work on other features we can think of and debug                             |
| Week 12<br>March 24 - 30 | Buffer / debugging             | Debugging/polishing           | Bugs      | Debug and prepare presentation                                               |

# Appendix A: Use Cases

## Subsection A: Route Manipulation

|                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 1                                                                                                                                                                                                                                                                                                                                                             |
| Title                   | Typical Maps User                                                                                                                                                                                                                                                                                                                                                      |
| Preparer                | Jessie Taubert                                                                                                                                                                                                                                                                                                                                                         |
| Actor/User              | Apple Maps / Google Maps / Waze User                                                                                                                                                                                                                                                                                                                                   |
| User Story              | My friend told me about this great app that uses AI to spice up my drives, so I downloaded it and want to use it as an alternative to the maps app I usually use.                                                                                                                                                                                                      |
| Course of Events        | <ol style="list-style-type: none"><li>1. The user creates an account to sign up</li><li>2. The user searches for their destination</li><li>3. The user selects their destination</li><li>4. The user selects the suggested route that takes the least time</li><li>5. The user selects “Go” button</li><li>6. The user selects the “Begin Navigation” button</li></ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                   |
| Related UI Sketch       | Sketch 15 -> Sketch 16 -> Sketch 1 -> Sketch 9 -> Sketch 7a                                                                                                                                                                                                                                                                                                            |

|                  |                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number           | Use Case 2                                                                                                                                                                                                                                                                                                                                                                               |
| Title            | Creating a Route From Scratch                                                                                                                                                                                                                                                                                                                                                            |
| Preparer         | Jessie Taubert                                                                                                                                                                                                                                                                                                                                                                           |
| Actor/User       | Driver                                                                                                                                                                                                                                                                                                                                                                                   |
| User Story       | Starting from my current location, I want to select the roads I am taking to a certain destination without any suggestions from the app.                                                                                                                                                                                                                                                 |
| Course of Events | <ol style="list-style-type: none"><li>1. The user clicks the search bar</li><li>2. The user inputs their start point and destination</li><li>3. The user selects the “Create custom route” button from the search dropdown menu</li><li>4. Starting from their start point, the user clicks a connecting road, which is highlighted and added to the total time to destination</li></ol> |

|                         |                                                                                                                                                                                                                                                                  |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <ol style="list-style-type: none"> <li>The user repeats this process until they either select their destination or end their route</li> <li>The user then saves this route or begins navigation</li> </ol>                                                       |
| Exceptions / Alternates | <ul style="list-style-type: none"> <li>The user tries to select a road that is closed, or a one-way, or cannot be taken (walking path, etc). An exception is raised and the path is not selected.</li> <li>User tries to reselect the same road twice</li> </ul> |
| Related UI Sketch       | Sketch 1 -> Sketch 9 -> Sketch 5a -> Sketch 7a -> Sketch 7b -> Sketch 13                                                                                                                                                                                         |

|                         |                                                                                                                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 3                                                                                                                                                                                                                                                                   |
| Title                   | Save a Route                                                                                                                                                                                                                                                                 |
| Preparer                | Kira Halls                                                                                                                                                                                                                                                                   |
| Actor/User              | Driver                                                                                                                                                                                                                                                                       |
| User Story              | The driver wants to save the route they just created for later use                                                                                                                                                                                                           |
| Course of Events        | <ol style="list-style-type: none"> <li>User is on the “Finalized Route” screen</li> <li>User selects the “Bookmark” button</li> <li>A popup with a text box shows up asking user to input a name for their route</li> <li>User inputs a name and clicks “Confirm”</li> </ol> |
| Exceptions / Alternates | User tried to name the route the same name as another saved route                                                                                                                                                                                                            |
| Related UI Sketch       | Sketch 7a -> Sketch 7b -> Sketch 13                                                                                                                                                                                                                                          |

|                  |                                                                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number           | Use Case 4                                                                                                                                                                                                     |
| Title            | Planning a Drive with Stops                                                                                                                                                                                    |
| Preparer         | Jessie Taubert                                                                                                                                                                                                 |
| Actor/User       | Driver / Navigator / Co Pilot                                                                                                                                                                                  |
| User Story       | We are going on a group road trip, and I want to plan out our path so we can stop at the best places for snacks and to use the restroom. I want to do this the night before and be ready to go in the morning! |
| Course of Events | <ol style="list-style-type: none"> <li>The user searches for their destination</li> <li>User selects the “Add stop” button</li> <li>The user searches for the added stop via the search bar</li> </ol>         |



|                         |                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <ol style="list-style-type: none"> <li>4. User selects a suggested route from the list</li> <li>5. User clicks “Go”</li> <li>6. The user saves this route by clicking the bookmark button</li> <li>7. Popup asks user to give the route a name before saving</li> <li>8. In the morning, the user goes to “Saved Routes” and selects the route</li> <li>9. The user selects “Begin Navigation”</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                                      |
| Related UI Sketch       | Sketch 1 -> Sketch 9 -> Sketch 7a -> Sketch 7b -> Sketch 13<br>Sketch 1 -> Sketch 3 -> Sketch 3a -> Sketch 7a -> Sketch 10a                                                                                                                                                                                                                                                                               |

|                         |                                                                                                                                                                                                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 5                                                                                                                                                                                                                                                                  |
| Title                   | Taking a previous route                                                                                                                                                                                                                                                     |
| Preparer                | Jessie Taubert                                                                                                                                                                                                                                                              |
| Actor/User              | Driver / Navigator / Co Pilot                                                                                                                                                                                                                                               |
| User Story              | I commute the same route to work every day and want an easy way to repopulate the route I take.                                                                                                                                                                             |
| Course of Events        | <ol style="list-style-type: none"> <li>1. The user opens the app.</li> <li>2. The user clicks “Preferences”</li> <li>3. The user selects “Saved Routes”</li> <li>4. The user clicks the route they want to take.</li> <li>5. The user selects “Begin Navigation”</li> </ol> |
| Exceptions / Alternates | The user could also be suggested, using previous history data, that they took a certain saved route at the same time the day before, and this suggestion could be populated at the home screen.                                                                             |
| Related UI Sketch       | Sketch 1 -> Sketch 3 -> Sketch 3a -> Sketch 7a -> Sketch 10a                                                                                                                                                                                                                |

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| Number     | Use Case 6                                                                      |
| Title      | Edit a Saved Route                                                              |
| Preparer   | Jessie Taubert                                                                  |
| Actor/User | Driver / Navigator / Co Pilot                                                   |
| User Story | Yesterday, I took a custom route to work, and it sent me down a dirt road and I |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | almost crashed! I want to avoid this road, and change that route!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Course of Events        | <ol style="list-style-type: none"> <li>1. The user clicks on “Preferences”</li> <li>2. The user clicks on “Saved Routes”</li> <li>3. User swipes right to reveal the “Edit” button</li> <li>4. The user selects “Edit”</li> <li>5. The user scrolls to the map displaying the path, and selects a new road to take.</li> <li>6. The user selects the green check mark to confirm their route</li> <li>7. The user selects the “Save route” button on the “Confirmed Route” screen</li> <li>8. A popup with a text box shows “Name of route” with the current name pre-populated</li> <li>9. User selects “Confirm”</li> <li>10. A popup asks “Do you want to add this road to “Avoided Roads”?”</li> <li>11. User changes the priority rating using the plus and minus buttons</li> <li>12. User selects “Yes”</li> </ol> |
| Exceptions / Alternates | <ul style="list-style-type: none"> <li>- User selects “No” on popup and road isn’t added to avoided roads</li> <li>- User edits the name of the route</li> <li>- User selects “Cancel” on the confirm route popup</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Related UI Sketch       | Sketch 1 -> Sketch 3a -> Sketch 5a -> Sketch 7a -> Sketch 7b -> Sketch 5b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Title                   | Make Small Change To Route                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Preparer                | Natalie Wilkins                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Actor/User              | Driver                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| User Story              | As a driver, I would like to be able to make a small change to my route without changing the entire route.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Course of Events        | <ol style="list-style-type: none"> <li>1. The user has already mapped a route and they can see it on the map.</li> <li>2. The user selects the pencil button indicating “Edit Route”</li> <li>3. The user selects the “Eraser” icon and clicks on the road they want to remove or</li> <li>4. The user selects the “Pencil” icon and clicks on the road they want to add to the route</li> <li>5. The route will remap, making the minimal amount of change possible.</li> <li>6. User selects the green confirmation button</li> <li>7. User selects “Begin Navigation” to be guided on the altered route</li> </ol> |
| Exceptions / Alternates | Route maps to one way / closed street                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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|-------------------|---------------------------------------------------|
| Related UI Sketch | Sketch 7a -> Sketch 5a -> Sketch 7a -> Sketch 10a |
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|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 8                                                                                                                                                                                                                                                                                                                                                                          |
| Title                   | Undo/Redo Change to Route                                                                                                                                                                                                                                                                                                                                                           |
| Preparer                | Kira Halls                                                                                                                                                                                                                                                                                                                                                                          |
| Actor/User              | Driver                                                                                                                                                                                                                                                                                                                                                                              |
| User Story              | The user has just added/deleted a road to their route that they did not mean to change. They would like to undo that change without too much hassle                                                                                                                                                                                                                                 |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User is already in the “Edit Route” screen</li> <li>2. User selects the “Undo” button and the route is restored to its previous state</li> <li>3. If the user realizes they actually did make the right change, they can select the “Redo” button to restore the route to the changed state</li> <li>4. User continues editing</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                |
| Related UI Sketch       | Sketch 7a                                                                                                                                                                                                                                                                                                                                                                           |

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|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number           | Use Case 9                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Title            | Take Route Without Highways                                                                                                                                                                                                                                                                                                                                                                                                       |
| Preparer         | Natalie Wilkins                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Actor/User       | Nervous driver                                                                                                                                                                                                                                                                                                                                                                                                                    |
| User Story       | As a nervous driver, I want to avoid highways and have the entirety of my route take place on surface roads.                                                                                                                                                                                                                                                                                                                      |
| Course of Events | <ol style="list-style-type: none"> <li>1. The user has entered a standard or custom route.</li> <li>2. The user clicks on the three dots symbol.</li> <li>3. The “Preferences” screen opens.</li> <li>4. The user selects the drop down menu for “Avoid”</li> <li>5. The user selects the checkbox next to “Avoid Highways”</li> <li>6. The user clicks outside of the “Preferences” menu to return to the main screen</li> </ol> |

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|-------------------------|-------------------------------------------------|
| Exceptions / Alternates | None                                            |
| Related UI Sketch       | Sketch 7a -> Sketch 3 -> Sketch 3b -> Sketch 7a |

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|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 10                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Title                   | Take Route Without Tolls                                                                                                                                                                                                                                                                                                                                                                                                       |
| Preparer                | Kira Halls                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Actor/User              | Financially Conscious Driver                                                                                                                                                                                                                                                                                                                                                                                                   |
| User Story              | As a nervous driver, I want to save money by avoiding tolls where possible                                                                                                                                                                                                                                                                                                                                                     |
| Course of Events        | <ol style="list-style-type: none"> <li>1. The user has entered a standard or custom route.</li> <li>2. The user clicks on the three dots symbol.</li> <li>3. The “Preferences” screen opens.</li> <li>4. The user selects the drop down menu for “Avoid”</li> <li>5. The user selects the checkbox next to “Avoid Tolls”</li> <li>6. The user clicks outside of the “Preferences” menu to return to the main screen</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Related UI Sketch       | Sketch 7a -> Sketch 3 -> Sketch 3b -> Sketch 7a                                                                                                                                                                                                                                                                                                                                                                                |

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|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number           | Use Case 11                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Title            | Turn Off Voice Navigation                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Preparer         | Natalie Wilkins                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Actor/User       | Driver                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| User Story       | As a driver, I want to see navigation information and listen to a playlist on the integrated player, but I don’t want navigation directions announced to me.                                                                                                                                                                                                                                                                                                        |
| Course of Events | <ol style="list-style-type: none"> <li>1. The user has entered a route and has either begun navigation or is still on the pre-navigation screen.</li> <li>2. The user clicks on the speaker button to toggle voice navigation.</li> <li>3. If voice navigation was previously on, it will now be off and show a line through the speaker symbol. If voice navigation was previously off, it will now be on and show a speaker symbol with no line. Voice</li> </ol> |

|                         |                                                                          |
|-------------------------|--------------------------------------------------------------------------|
|                         | navigation is on by default.                                             |
| Exceptions / Alternates | The user's phone volume is down and they try to turn on voice navigation |
| Related UI Sketch       | Sketch 7a -> Sketch 10b                                                  |

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|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 12                                                                                                                                                                                                                                                                                                                  |
| Title                   | Begin Navigation                                                                                                                                                                                                                                                                                                             |
| Preparer                | Natalie Wilkins                                                                                                                                                                                                                                                                                                              |
| Actor/User              | Driver                                                                                                                                                                                                                                                                                                                       |
| User Story              | As a user, after I have input my custom route, I want to indicate that I am beginning to drive.                                                                                                                                                                                                                              |
| Course of Events        | <ol style="list-style-type: none"> <li>1. The user has entered their route.</li> <li>2. The user clicks the "Begin Navigation" button.</li> <li>3. The app will change to Navigation Mode, indicating to the user that it will begin to give verbal directions and indicate the next turn and the distance to it.</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                         |
| Related UI Sketch       | Sketch 7a -> Sketch 10a                                                                                                                                                                                                                                                                                                      |

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|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number           | Use Case 13                                                                                                                                                                                      |
| Title            | Ending Navigation                                                                                                                                                                                |
| Preparer         | Kira Halls                                                                                                                                                                                       |
| Actor/User       | Driver                                                                                                                                                                                           |
| User Story       | Halfway through a drive, a driver realizes they no longer need navigation assistance so they want to end the route they are taking                                                               |
| Course of Events | <ol style="list-style-type: none"> <li>1. User is already on the "Navigation" page</li> <li>2. User selects "End Route"</li> <li>3. Navigation ends and user is returned to Main Page</li> </ol> |

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|-------------------------|------------------------|
| Exceptions / Alternates | None                   |
| Related UI Sketch       | Sketch 10a -> Sketch 1 |

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| Number                  | Use Case 14                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Title                   | Adding Stop during Navigation                                                                                                                                                                                                                                                                                                                                                                                               |
| Preparer                | Kira Halls                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Actor/User              | Driver                                                                                                                                                                                                                                                                                                                                                                                                                      |
| User Story              | While driving along a route, the driver realizes they need to stop for gas but they need some help navigating to a gas station, so they need to add a stop                                                                                                                                                                                                                                                                  |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User is already on the “Navigation” page</li> <li>2. The user selects “Add stop”</li> <li>3. The user searches for their stop</li> <li>4. The user clicks “Confirm”</li> <li>5. The app remaps the route and navigates them to the gas station</li> <li>6. Once the user leaves the gas station they are navigated along their original route without any extra action</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Related UI Sketch       | Sketch 10a -> Sketch 10c                                                                                                                                                                                                                                                                                                                                                                                                    |

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|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number           | Use Case 15                                                                                                                                                                                                                                 |
| Title            | Sending a route                                                                                                                                                                                                                             |
| Preparer         | Jessie Taubert                                                                                                                                                                                                                              |
| Actor/User       | Parent / Wisdom Holder                                                                                                                                                                                                                      |
| User Story       | I am a parent and know the best way to get to a certain place. My child just started driving and I want to send them the route that they should take.                                                                                       |
| Course of Events | <ol style="list-style-type: none"> <li>1. The parent creates the route in the app</li> <li>2. The parent clicks “Share”</li> <li>3. The parent clicks “Messages”</li> <li>4. The parent sends the route to their child’s contact</li> </ol> |

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|                         | <ol style="list-style-type: none"> <li>The child clicks on the link</li> <li>The child begins navigation on the route, or saves it</li> </ol> |
| Exceptions / Alternates | Message fails to send so a “Message failed” error is raised                                                                                   |
| Related UI Sketch       | Sketch 7a -> Sketch 14                                                                                                                        |

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|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 16                                                                                                                                                                                                                                                                                                                |
| Title                   | Report Hazard or Traffic Accident                                                                                                                                                                                                                                                                                          |
| Preparer                | Natalie Wilkins                                                                                                                                                                                                                                                                                                            |
| Actor/User              | Driver                                                                                                                                                                                                                                                                                                                     |
| User Story              | As a driver, I want to be able to report a hazard or traffic accident so other users can avoid it.                                                                                                                                                                                                                         |
| Course of Events        | <ol style="list-style-type: none"> <li>The user is driving when they encounter a hazard or accident.</li> <li>The user clicks on the “Report Hazard / Accident” button</li> <li>The user selects the type of incident and its location and submits the report.</li> <li>The app displays a confirmation screen.</li> </ol> |
| Exceptions / Alternates | <ul style="list-style-type: none"> <li>The user doesn’t select a type of incident or location</li> <li>Network connection is lost so report doesn’t submit and confirmation is never given</li> </ul>                                                                                                                      |
| Related UI Sketch       | Sketch 10a -> Sketch 11                                                                                                                                                                                                                                                                                                    |

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|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number           | Use Case 17                                                                                                                                                                                                                                               |
| Title            | Enjoyment of ride                                                                                                                                                                                                                                         |
| Preparer         | Nate Stovak                                                                                                                                                                                                                                               |
| Actor/User       | Any user                                                                                                                                                                                                                                                  |
| User Story       | The user takes a route that they have not taken before, or one that they take frequently but have not rated recently. The user is then given a pop-up after the ride so they can rate their enjoyment of the ride so the app can recommend similar routes |
| Course of Events | <ol style="list-style-type: none"> <li>The user ends their route</li> <li>A pop-up in the app comes up with stars for the user to rate the route that they are on out of 5 stars</li> <li>That route is then stored in their rated routes</li> </ol>      |

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|-------------------------|----------------------------------------|
| Exceptions / Alternates | Rating not saved to database correctly |
| Related UI Sketch       | Sketch 12                              |

#### Subsection B: Preferences Menu

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|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Title                   | Walking Mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Preparer                | Kira Halls                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Actor/User              | Walking User with physical disability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| User Story              | I am a user who has a physical disability that makes steep inclines hard to navigate. I want to use this app to guide me how to walk to my destination but I want to see areas with steep inclines that I need to avoid                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User selects the search bar</li> <li>2. User selects the “Walking” icon from the tab before they select a route or the “Create Custom Route” button</li> <li>3. User selects desired route</li> <li>4. User selects “Confirm Route”</li> <li>5. Steep inclines on the route are marked in red on the route, orange for medium incline, and green for low incline</li> <li>6. User selects the “Edit” button</li> <li>7. User selects the checkbox next to “avoid steep incline” label and “avoid medium incline” label to have a route suggested that only uses low incline roads</li> <li>8. The user clicks “Begin navigation” button</li> </ol> |
| Exceptions / Alternates | <ul style="list-style-type: none"> <li>- There is no suggested route that avoids red and orange levels of incline</li> <li>- Finding a fully green incline route adds a lot of mileage to the route</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Related UI Sketch       | Sketch 9 -> Sketch 7b -> Sketch 6 -> Sketch 7b -> Sketch 10a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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|------------|---------------------------------------------------------------------------|
| Number     | Use Case 19                                                               |
| Title      | Delete a Saved Route                                                      |
| Preparer   | Kira Halls                                                                |
| Actor/User | Driver                                                                    |
| User Story | I recently moved apartments and I no longer need my saved route from work |



|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | to my old apartment. I want to delete it since I won't be using it anymore                                                                                                                                                                                                                                                                                                                                                                                   |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User selects "Preferences"</li> <li>2. User selects "Saved Routes"</li> <li>3. User scrolls to the route they want to delete</li> <li>4. User swipes left to reveal a "Delete" button</li> <li>5. User selects "Delete"</li> <li>6. User sees a "Confirmation" popup that asks "Do you really want to delete this custom route?" and selects "Yes"</li> <li>7. Route is deleted from "Saved Route" list</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Related UI Sketch       | Sketch 1 -> Sketch 3 -> Sketch 3a                                                                                                                                                                                                                                                                                                                                                                                                                            |

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|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 20                                                                                                                                                                                                                                                                                                                                                                      |
| Title                   | App Customization                                                                                                                                                                                                                                                                                                                                                                |
| Preparer                | Kira Halls                                                                                                                                                                                                                                                                                                                                                                       |
| Actor/User              | Driver/Walker                                                                                                                                                                                                                                                                                                                                                                    |
| User Story              | I'm using the app for driving navigation or walking navigation and I hate the default map theme. I want to see a new map theme when I follow my route                                                                                                                                                                                                                            |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User selects the "Preferences" button</li> <li>1. User selects "Map Customization" from the preferences menu</li> <li>2. User selects a map theme from the dropdown list (dark mode, street view, satellite, topo, positron, bright, or basic)</li> <li>3. User's map experience is now customized with that selected theme</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                             |
| Related UI Sketch       | Sketch 1 -> Sketch 3 -> Sketch 3e                                                                                                                                                                                                                                                                                                                                                |

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|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number     | Use Case 21                                                                                                                                                 |
| Title      | Avoid routes                                                                                                                                                |
| Preparer   | Nate Stovak                                                                                                                                                 |
| Actor/User | Gym-goer                                                                                                                                                    |
| User Story | A university student loves to go to the gym. He goes up 100 S but hates how many potholes it has and how close he is to the students on the sidewalk and is |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | looking to avoid that road as it makes him anxious for the students, himself, and his car.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User selects the “Preference” button</li> <li>2. User selects “Avoid”</li> <li>3. User selects “Roads to Avoid”</li> <li>4. User clicks the “Add Road” button at the bottom of the menu</li> <li>5. User types in 100 S in the search bar and presses drop down showing users how much they would like to avoid that road a number 1-3.</li> <li>6. “Confirmation” screen pops up asking user if they would like to avoid 100 S</li> <li>7. User confirms and that route is avoided based on their number provided</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Related UI Sketch       | Sketch 1 -> Sketch 3 -> Sketch 3b -> Sketch 3c -> Sketch 3d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 22                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Title                   | Remove avoided road                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Preparer                | Nate Stovak                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Actor/User              | Gym-goer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| User Story              | A university student loves to go to the gym. He previously avoided 100 S because it contained massive potholes, however the road was recently paved. Since his displeasure with the route was solved he removes it from his avoid roads                                                                                                                                                                                                                                                   |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User selects the “Preference button”</li> <li>2. User selects “Roads to Avoid”</li> <li>3. User swipes left on the “100 S” button under the “Roads to Avoid” column</li> <li>4. User selects the “Delete” button that pops up</li> <li>5. “Confirmation” screen pops up asking user if they would like to remove 100 S from their “Roads to Avoid”</li> <li>8. User confirms and that route is now seen as a normal route by the app</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Related UI Sketch       | Sketch 1 -> Sketch 3 -> Sketch 3b -> Sketch 3c                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| Number | Use Case 23 |
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|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title                   | Receive A Playlist Recommendation                                                                                                                                                                                                                                                        |
| Preparer                | Natalie Wilkins                                                                                                                                                                                                                                                                          |
| Actor/User              | Driver                                                                                                                                                                                                                                                                                   |
| User Story              | As a driver, I want to be given a recommended playlist and be able to play, pause, skip within CustomMaps.                                                                                                                                                                               |
| Course of Events        | <ol style="list-style-type: none"> <li>1. The user clicks on the “Preferences” button.</li> <li>2. The user selects “Playlists”.</li> <li>3. The user selects “Recommend Something”.</li> <li>4. The music begins playing and has an integrated player in the CustomMaps app.</li> </ol> |
| Exceptions / Alternates | <ul style="list-style-type: none"> <li>- Network error</li> <li>- API error</li> </ul>                                                                                                                                                                                                   |
| Related UI Sketch       | Sketch 1 -> Sketch 3 -> Sketch 3g -> Sketch 4                                                                                                                                                                                                                                            |

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|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 24                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Title                   | Tree-based route                                                                                                                                                                                                                                                                                                                                                                                                            |
| Preparer                | Nate Stovak                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Actor/User              | User looking for a different way home                                                                                                                                                                                                                                                                                                                                                                                       |
| User Story              | A user lives in a very hot city such as Salt Lake during Summer and has no air conditioning in their car. As a result they are burning up in their car and would like to find a way home that provides lots of shade, so they use our app’s tree setting. This setting provides a route to the destination that includes roads with lots of trees and shade.                                                                |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User clicks the “Preference” button</li> <li>2. User clicks the “Scenic Preferences” button</li> <li>3. User then clicks the “Trees” preference and inputs from one to three using the plus and minus buttons how much deviation from the most efficient route they would go</li> <li>4. Then the route is created and allows the user to change it as they would like</li> </ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Related UI Sketch       | Sketch 1 -> Sketch 3 -> Sketch 3f -> Sketch 9                                                                                                                                                                                                                                                                                                                                                                               |

### Subsection C: Settings Menu

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|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Title                   | Avoid tracking data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Preparer                | Nate Stovak                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Actor/User              | Worried civilian                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| User Story              | A civilian likes using map applications to get around the city, but does not like it when it saves and uses their data in real-time and afterwards. The application has a toggle switch that allows the user to ensure that their data is not being saved at any time. This would mean no saved routes, avoided roads, and any other additional features requiring or storing data.                                                                                                                        |
| Course of Events        | <ol style="list-style-type: none"><li>1. The user selects the “Settings” button</li><li>2. User switches the toggle button saying “Store User Data for personalized experience”</li><li>3. Confirmation pop-up comes up saying that all previous data for the user will be deleted. Along with the user not being able to use the portions of the app that require saved data</li><li>4. User presses yes, and all data is deleted and all portions of the app that require data are gray-scaled</li></ol> |
| Exceptions / Alternates | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Related UI Sketch       | Sketch 1 -> Sketch 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

### Subsection D: Login

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|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number           | Use Case 26                                                                                                                                                                                |
| Title            | Login Screen                                                                                                                                                                               |
| Preparer         | Nate Stovak                                                                                                                                                                                |
| Actor/User       | All users                                                                                                                                                                                  |
| User Story       | A login screen will appear for the user to use our application so we can understand who is using the application. The user can either create an account or input their email and password. |
| Course of Events | <ol style="list-style-type: none"><li>1. User opens app</li></ol>                                                                                                                          |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <ol style="list-style-type: none"> <li>2. User is given login screen with options to put in credentials, create account, recover password, and stay logged in</li> <li>3. User inputs their email in the “Email” field</li> <li>4. User inputs their password in the “Password” field</li> <li>5. User selects the “Remember me” checkbox to stay logged in the next time they open the app</li> <li>6. User clicks “Log in”</li> </ol> |
| Exceptions / Alternates | <ul style="list-style-type: none"> <li>- User tries an email that doesn’t exist in our system–”Account does not exist” exception is raised</li> <li>- User inputs incorrect password–”Incorrect password” exception is raised</li> </ul>                                                                                                                                                                                                |
| Related UI Sketch       | Sketch 15 -> Sketch 1                                                                                                                                                                                                                                                                                                                                                                                                                   |

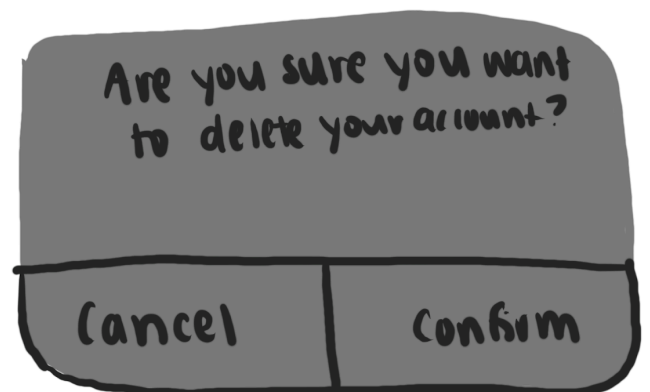
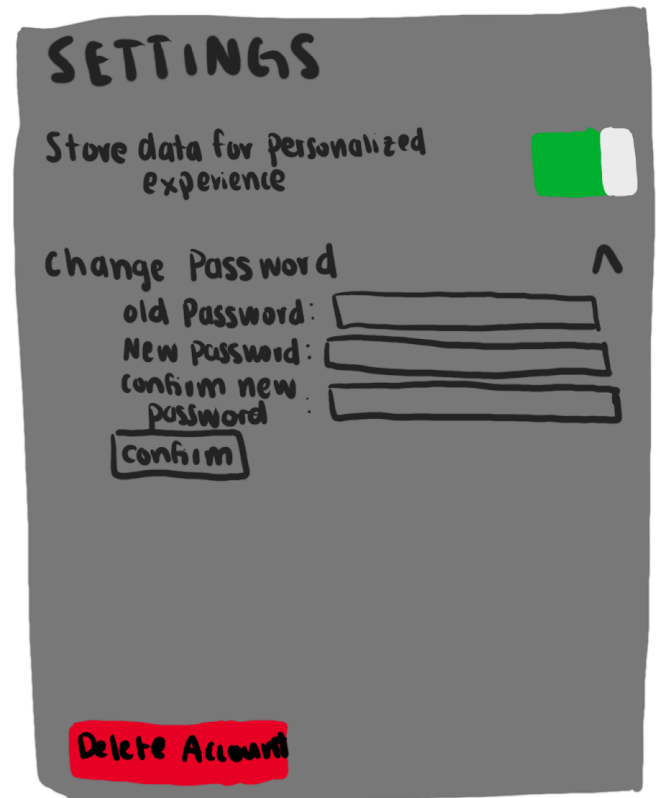
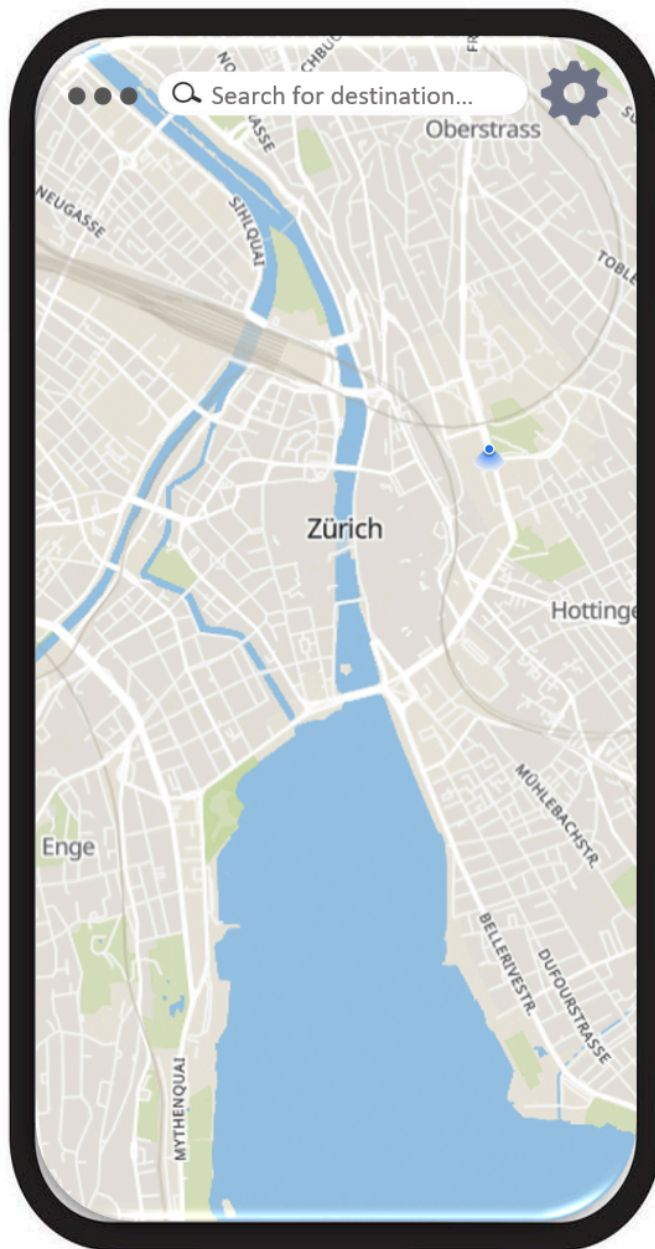
|                         |                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 27                                                                                                                                                                                                                                                                                                                                                         |
| Title                   | Create an account                                                                                                                                                                                                                                                                                                                                                   |
| Preparer                | Natalie Wilkins                                                                                                                                                                                                                                                                                                                                                     |
| Actor/User              | All users                                                                                                                                                                                                                                                                                                                                                           |
| User Story              | As a user, I want to be able to create an account where I can save my routes. I want to be able to create an account from my email and a password.                                                                                                                                                                                                                  |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User selects the “Create one here” link at the bottom of the login page</li> <li>2. User inputs their email in the “Email” field</li> <li>3. User inputs their password in the “Password” field</li> <li>4. User retypes password in the “Confirm Password” field</li> <li>5. User selects the “Create” button</li> </ol> |
| Exceptions / Alternates | <ul style="list-style-type: none"> <li>- User inputs an invalid email format so an “Invalid Format” exception is raised</li> <li>- The “password” and “confirm password” fields don’t match so “Fields don’t match” exception is raised</li> </ul>                                                                                                                  |
| Related UI Sketch       | Sketch 15 -> Sketch 16                                                                                                                                                                                                                                                                                                                                              |

|          |                 |
|----------|-----------------|
| Number   | Use Case 28     |
| Title    | Change Password |
| Preparer | Kira Halls      |

|                         |                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actor/User              | App User                                                                                                                                                                                                                                                                                                                                                   |
| User Story              | I am worried my current account password is too easy to guess. I want to change it to be more secure                                                                                                                                                                                                                                                       |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User opens app</li> <li>2. User clicks the “Settings” button</li> <li>3. User clicks “Change password”</li> <li>4. User inputs current password</li> <li>5. User inputs new password in the “New Password” field and retypes it in the “Confirm new password” field</li> <li>6. User clicks “Confirm”</li> </ol> |
| Exceptions / Alternates | <ul style="list-style-type: none"> <li>- The “New Password” and “Confirm New Passwords” field do not match so a “Password and confirmation do not match” error is raised</li> <li>- User types in the wrong current password so an “Incorrect current password” exception is raised</li> </ul>                                                             |
| Related UI Sketch       | Sketch 1 -> Sketch 2                                                                                                                                                                                                                                                                                                                                       |

|                         |                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number                  | Use Case 29                                                                                                                                                                                                                                                                                                                                                                     |
| Title                   | Delete Account                                                                                                                                                                                                                                                                                                                                                                  |
| Preparer                | Kira Halls                                                                                                                                                                                                                                                                                                                                                                      |
| Actor/User              | Former User                                                                                                                                                                                                                                                                                                                                                                     |
| User Story              | I no longer need to use the CustomMaps app, and I want to delete my account so my data isn’t being stored                                                                                                                                                                                                                                                                       |
| Course of Events        | <ol style="list-style-type: none"> <li>1. User opens app</li> <li>2. User clicks on the “Settings” button</li> <li>3. User clicks “Delete Account” button at the bottom of the settings page</li> <li>4. A popup shows up asking “Are you sure you want to delete your account?”</li> <li>5. User selects “Confirm”</li> <li>6. User is directed to the “Login” page</li> </ol> |
| Exceptions / Alternates | The user’s data is not successfully deleted from the database                                                                                                                                                                                                                                                                                                                   |
| Related UI Sketch       | Sketch 1 -> Sketch 2 -> Sketch 2b                                                                                                                                                                                                                                                                                                                                               |

## Appendix B: UI Sketches



| Preferences        |   |
|--------------------|---|
| Saved Routes       | > |
| Avoid              | ✓ |
| Map theme          | ✓ |
| Scenic Preferences | ✓ |
| Playlists          | ✓ |

| Preferences                                                                                |   |
|--------------------------------------------------------------------------------------------|---|
| Saved Routes                                                                               | > |
| Avoid                                                                                      | ^ |
| <input checked="" type="checkbox"/> Avoid Highways<br><input type="checkbox"/> Avoid Tolls |   |
| Roads to Avoid                                                                             | > |
| Map theme                                                                                  | ✓ |
| Scenic preferences                                                                         | ✓ |
| Playlists                                                                                  | ✓ |

|                       |                 |
|-----------------------|-----------------|
| Work - 20 min         |                 |
| Delete                | School - 30 min |
| Grocery Store - 5 min |                 |
| Edit                  |                 |

|                 |                |           |
|-----------------|----------------|-----------|
| 100 South - SLC |                | Rating: 2 |
| Delete          | 400 East - SLC | Rating: 1 |
| + Add Road      |                |           |



 Search for road

How much do you want to avoid? ✓

1 - Low priority

2 - Medium priority

3 - High priority

Cancel

Confirm

Preferences

Saved Routes >

Avoid ✓

Map theme ^

○ Streets view

○ Satellite view

○ Topo view

○ Basic

○ Bright

○ Positron

● Dark Matter

Scenic preferences ✓

Playlists ✓

Preferences

Saved Routes >

Avoid ✓

Map theme ✓

Scenic Preferences ^

☐ Historical Sites

☐ Flowers

☒ Trees -2+

Playlists ✓

Preferences

Saved Routes >

Avoid ✓

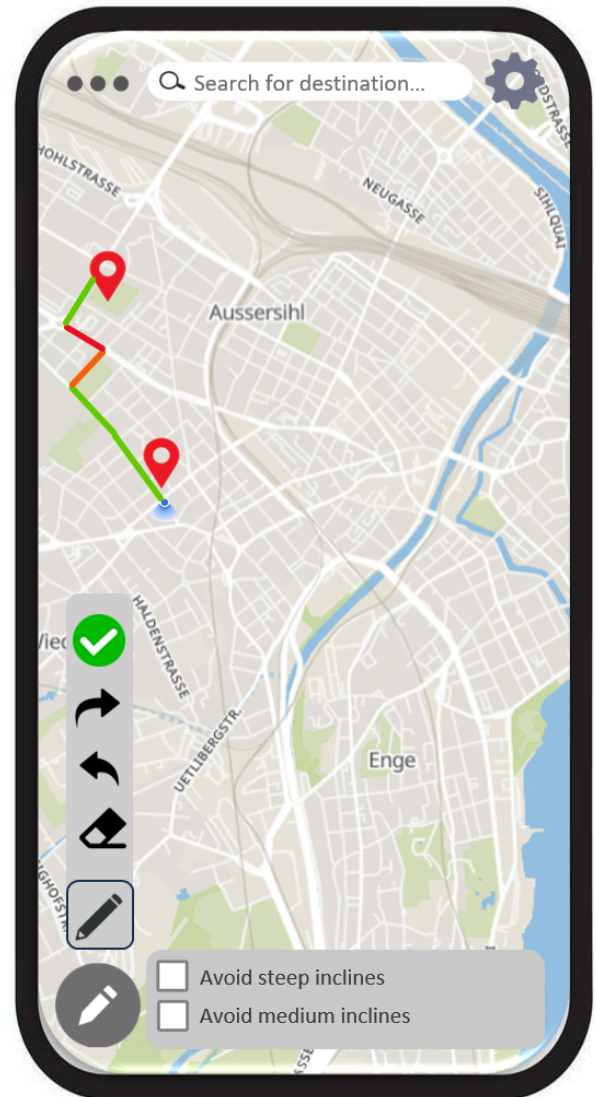
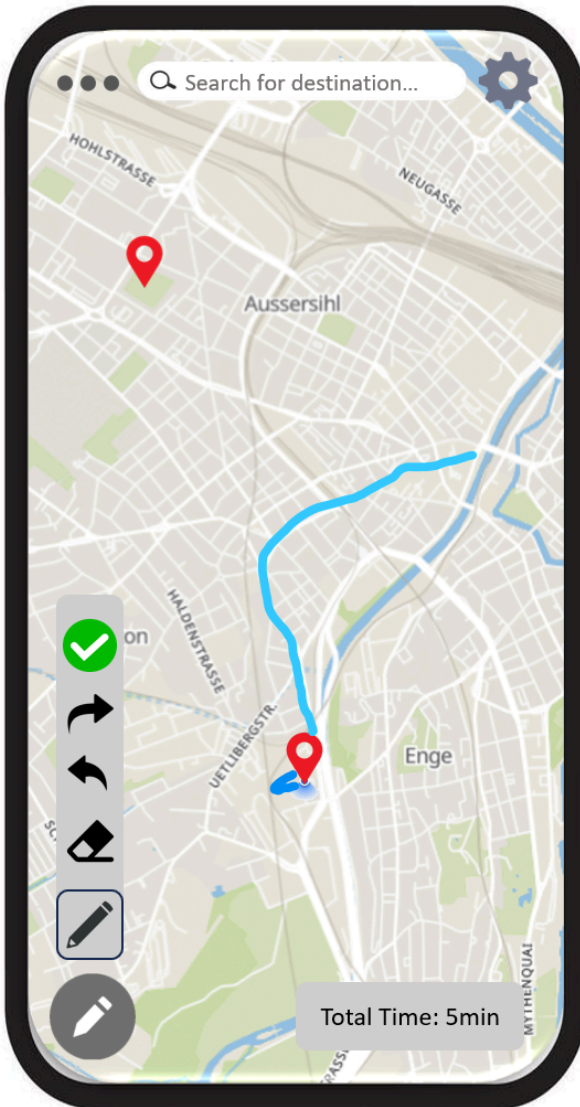
Map theme ✓

Scenic Preferences ✓

Playlists ^

Recommend me something

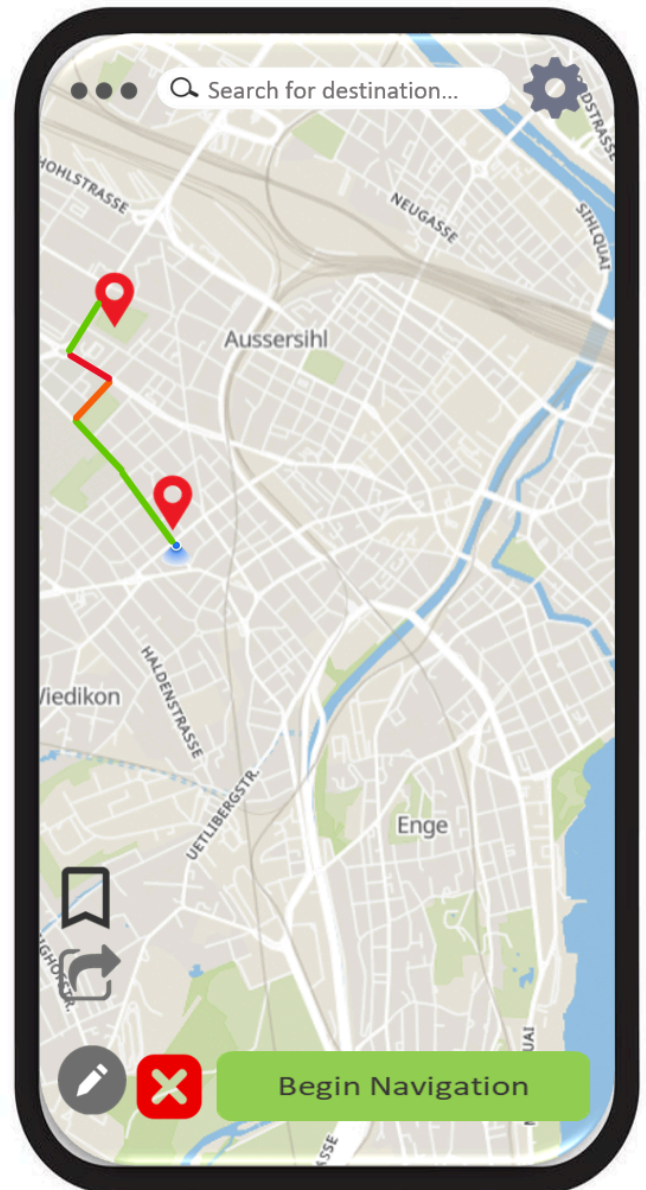
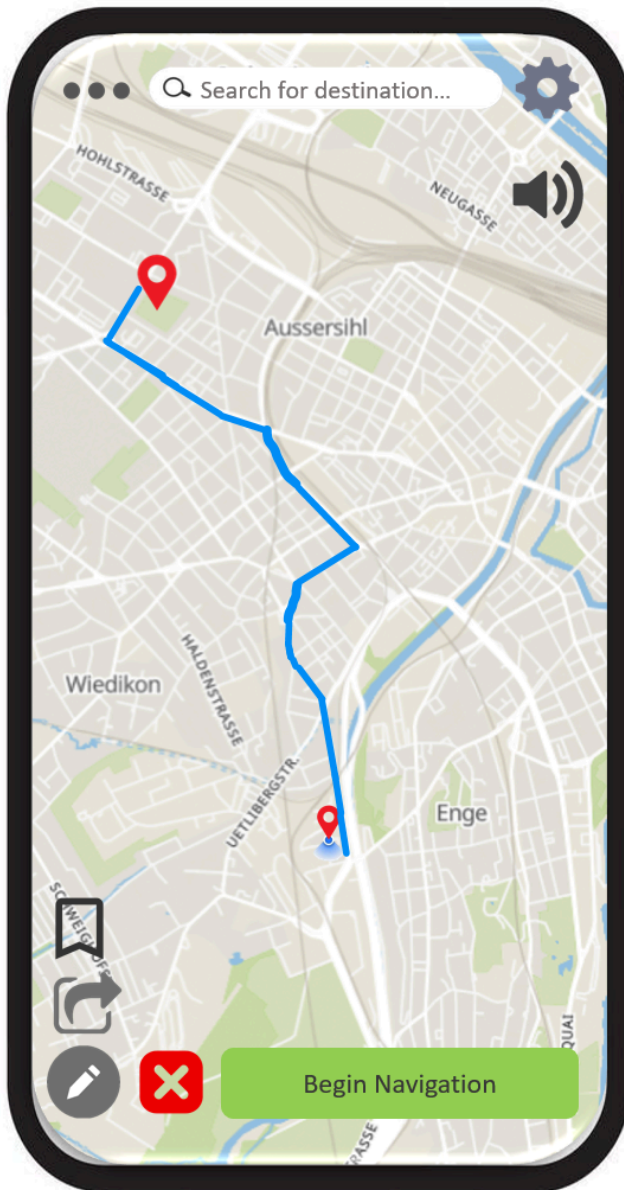
Choose my playlist ≡



Do you want to add  
this road to  
"Avoided Roads"?

Priority: -1 +

|    |     |
|----|-----|
| No | Yes |
|----|-----|



From...

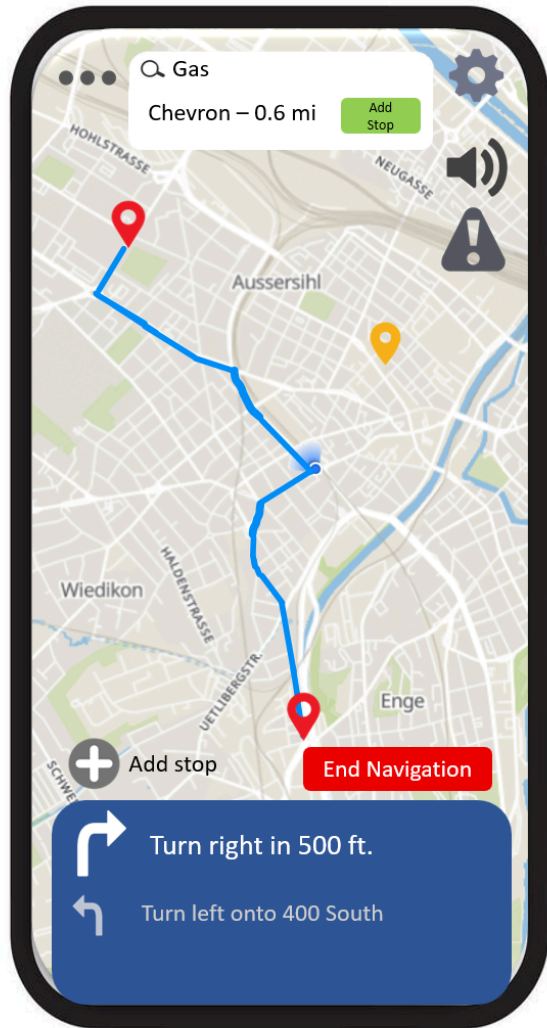
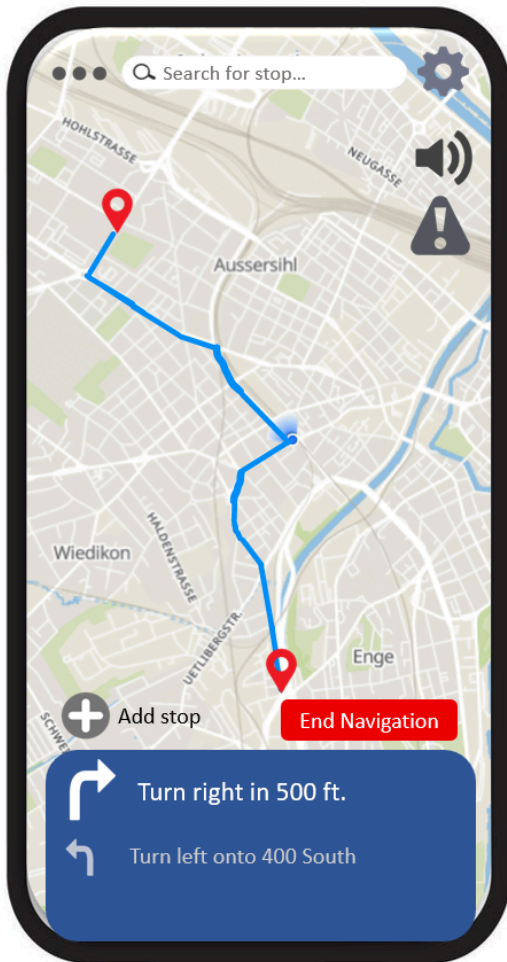
+ Add Stop

Destination...

|                               |    |
|-------------------------------|----|
| 17 min — 10 mi                | GO |
| 20 min — 7.7 mi               | GO |
| 20 min — 10 mi —  Trees Route | GO |

Create custom route



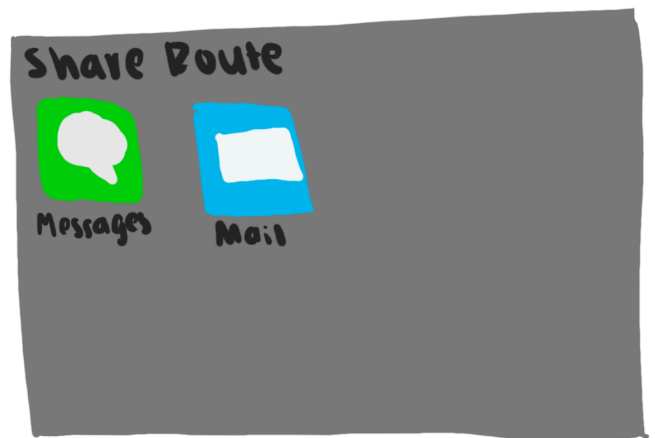
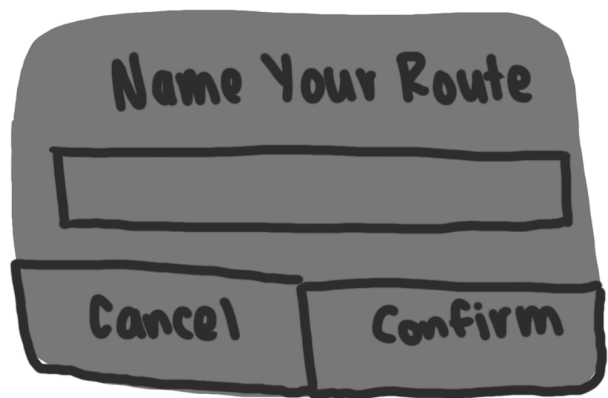
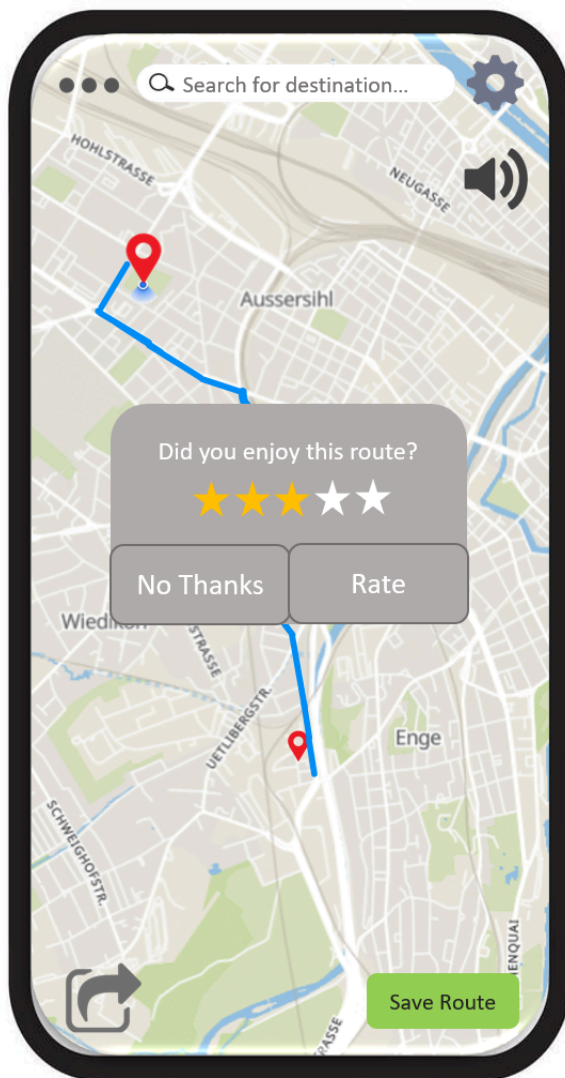


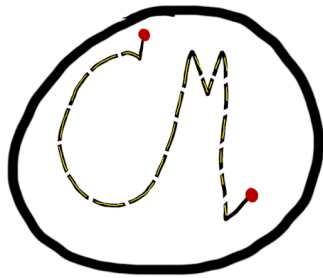
## Report Incident

Incident Type... ▾

Q location...

Cancel | Report





## LOG IN

Email:

Password:

☒ Remember Me [Forgot Password?](#)

LOG IN

---

Log in with Google G

Don't have an account? [Create one here](#)

## CREATE ACCOUNT

Email:

Password:

Confirm Password:

Create

# Revisions Appendix

## 1. Executive Summary

- a. Our executive summary remains mostly the same, but we updated the technologies we are planning to use.

## 2. Background

- a. The background research remains the same. The only change was to the technologies section, reflecting our changes from Flutter and Dart to JavaScript and React Native.

## 3. Requirements Analysis

- a. We changed the system architecture diagram to better reflect the system architecture explanation.
- b. We also changed the system architecture explanation to include the tech stack changes.
- c. We added a feature code to label our requirements as per the suggestion.
- d. We significantly shrunk the information in this section down as much as we could without reducing the detail in our documentation.

## 4. Software Engineering Tools and Techniques

- a. We changed our tech stack a bit, from Flutter and Dart to JavaScript and React Native with Expo and Node.js, as well as from OpenMapTiles to Google Cloud APIs.

## 5. Timeline

- a. No changes were made to this section, as it hasn't had time to be iterated upon.

## 6. Appendix A: Use Cases

- a. No changes were made to this section.

## 7. Appendix B: UI Sketches

- a. No changes were made to the sketches.